

QALG - Quantum Algorithms

QCE26 Techical Papers Track

v123	Day	Time	Room	Track	QCE-ID	Session Name	ECID	Paper Authors	Paper Title
1	0-SUN	10:00 -11:30	713B	QALG	QALG::222::504::767	State Preparation & Oracles	222	Yusuke Kimura and Yutaka Takita	Database Reordering for Compact Grover Oracles with ESOP Minimization
2	0-SUN	10:00 -11:30	713B	QALG	QALG::222::504::767	State Preparation & Oracles	504	Debora Ramacciotti, Martin Steinbach, Bence Temesi, Andreea Iulia Lefterovici and Antonio F. Rotundo	Approximate sparse state preparation with Grover-Rudolph algorithm
3	0-SUN	10:00 -11:30	713B	QALG	QALG::222::504::767	State Preparation & Oracles	767	Nicholas Zhao	Ancilla-Efficient \textsc{QSample} Preparation for Reversible Markov Chains
4	0-SUN	13:00 -14:30	713B	QALG	QALG::449::616::661	Quantum Networking & Cryptography	449	Evan Anderson, Kaushik Datta and Boulat Bash	Toward Covert Quantum Computing
5	0-SUN	13:00 -14:30	713B	QALG	QALG::449::616::661	Quantum Networking & Cryptography	616	Varin Sikand and Andrew Nemec	Quantum Anonymous Secret Sharing with Permutation Invariant Codes
6	0-SUN	13:00 -14:30	713B	QALG	QALG::449::616::661	Quantum Networking & Cryptography	661	Carolyn Mayer, Anand Ganti and Uzoma Onunkwo	Enabling High-Speed Fusion-Based Quantum Computation via On-Demand Resource States
7	0-SUN	15:00 -16:30	713B	QALG	QALG::399::804::915	Quantum Machine Learning & Data	399	Luke James Miller and Yugyung Lee	QulC: A Training-Free Quantum Graph Embedding from Ideal Analysis to Practical Hardware Evaluation
8	0-SUN	15:00 -16:30	713B	QALG	QALG::399::804::915	Quantum Machine Learning & Data	804	Wentao Huang and Fanfan Lin	Quantum-Enhanced Knowledge Graph Systems: Survey, Taxonomy, and the Road to Scalability
9	0-SUN	15:00 -16:30	713B	QALG	QALG::399::804::915	Quantum Machine Learning & Data	915	Lakshikka Sithamparanathan and Sathish Kumar	Spectral Learning Rates for Bandwidth-Tuned Quantum Kernels
10	1-MON	10:00 -11:30	713B	QALG	QALG::471::541::630	Simulation & Scientific Computing	471	Erico Souza Teixeira, Amanda Marques de Lima, Larissa de Araújo Bourbon, Eivson Darlivam Rodrigues de Aguiar Silva, Eduardo Padrón Hernández, Daniel Felinto Pires Barbosa and Ricardo Luiz Longo	Active-Space VQE with Symmetry-Conserving Bravyi-Kitaev Mapping: Efficient Reference State Preparation and Compact Ansatz Operators
11	1-MON	10:00 -11:30	713B	QALG	QALG::471::541::630	Simulation & Scientific Computing	541	Brian Goldsmith, Larissa Kroell and Nishna Aerabati	Quantum Resource Estimation for Simulating the SYK Model with Trotterization, qDRIFT, and Asymmetric Qubitization
12	1-MON	10:00 -11:30	713B	QALG	QALG::471::541::630	Simulation & Scientific Computing	630	Chao Lu, Muralikrishnan Gopalakrishnan Meena, Eduardo Antonio Coello Pérez, Kalyana Gottiparthi and Seongmin Kim	Scaling Variational Quantum Linear Solver on a Hybrid Quantum-HPC Stack
13	1-MON	10:00 -11:30	713A	QALG	QALG::408::1143	QAOA for Optimization Problems	408	Bao Bach, Filip Maciejewski and Ilya Safro	Warm-Starting Rank-2 MaxCut Relaxation Using Local Correlators From Depth-1 Quantum Approximate Optimization Algorithm
14	1-MON	10:00 -11:30	713A	QALG	QALG::408::1143	QAOA for Optimization Problems	1143	Kieran McDowall, Konstantinos Georgopoulos and Petros Wallden	A Spectral Gap Informed Parameter Schedule for QAOA
16	1-MON	13:00 -14:30	713B	QALG	QALG::1199::1220::1246	Quantum LDPC & Decoder Design	1199	Milad Taghipour, Dimitris Chytas and Bane Vasic	Multistage Rewinding Decoder for QLDPC Codes
17	1-MON	13:00 -14:30	713B	QALG	QALG::1199::1220::1246	Quantum LDPC & Decoder Design	1220	Dimitris Chytas, Paul Fessatidis, Boulat Bash and Bane Vasic	Edge-Based Anisotropic Decoding for Generalized Bicycle Codes
18	1-MON	13:00 -14:30	713B	QALG	QALG::1199::1220::1246	Quantum LDPC & Decoder Design	1246	Shobhit Bhatnagar, Michele Pacenti, Nithin Raveendran, David Declercq and Bane Vaisc	Impulse Decoding: A Code-Shortening Perspective on Degeneracy in Quantum Error-Correcting Codes
19	1-MON	13:00 -14:30	713A	QALG	QALG::413::232::491	Constrained Combinatorial Optimization I	413	Tristan Zaborniak, Prashanti Priya Angara, Vikram Khipple Mulligan, Hausi Müller and Ulrike Stege	Qubit-efficient and gate-efficient encodings of graph partitioning problems for quantum optimization
20	1-MON	13:00 -14:30	713A	QALG	QALG::413::232::491	Constrained Combinatorial Optimization I	232	Fynn Meyer, Leon Froelje, Lennart Weingarten, Kamalika Datta, Abhoy Kole, Christoph Lueth and Rolf Drechsler	An Efficient Encoding for Subset Sum Problem Exploiting QFT-Based Arithmetic Operators
21	1-MON	13:00 -14:30	713A	QALG	QALG::413::232::491	Constrained Combinatorial Optimization I	491	Lennart Binkowski, Tobias J. Osborne, Marvin Schwiering, René Schwonnek and Timo Ziegler	One for All: A Universal Quantum Conic Programming Framework for Hard-Constrained Combinatorial Optimisation Problems
22	1-MON	15:00 -16:30	713B	QALG	QALG::635::875::910	Error Detection & Mitigation II	635	Noble Agyeman-Bobie, Quinn Langfitt, Salahdeen Issa, Nikos Hardavellas and Kaitlin Smith	Generalizing Pauli Checks for Qudit-based Quantum Error Detection and Mitigation
23	1-MON	15:00 -16:30	713B	QALG	QALG::635::875::910	Error Detection & Mitigation II	875	Juan Francisco Martín Bravo, Giuseppe Cocco and Javier Rodríguez Fonollosa	Quantum Error Mitigation with a Surrogate Observable
24	1-MON	15:00 -16:30	713B	QALG	QALG::635::875::910	Error Detection & Mitigation II	910	Rohan Kumar, Ben Foxman and Yongshan Ding	Modeling Quantum Error Detection with Transition Matrices
25	1-MON	15:00 -16:30	713A	QALG	QALG::332::647::391	Constrained Combinatorial Optimization II	332	Loong Kuan Lee, Harsha Nagarajan, Thore Gerlach, Sascha Mücke, Ragavi Krishnamoorthy and Nico Piatkowski	Hardware-aware QUBO Reformulation of Constrained Binary Optimization via the Walsh–Fourier Transform
26	1-MON	15:00 -16:30	713A	QALG	QALG::332::647::391	Constrained Combinatorial Optimization II	647	Marvin Schwiering, Timo Ziegler, Lennart Binkowski and Benjamin Sambale	Exhaustive and feasible parametrisation with applications to the travelling salesperson problem
27	1-MON	15:00 -16:30	713A	QALG	QALG::332::647::391	Constrained Combinatorial Optimization II	391	Sören Wilkening, Timo Ziegler and Maximilian Hess	Constraint-oriented biased quantum search for linear constrained combinatorial optimization problems
28	2-TUE	10:00 -11:30	713B	QALG	QALG::921::949::1255	Quantum Information & Miscellaneous	921	Sara Santos, Stefan Woerner, Vincenzo Savona and Julien Gacon	Efficient re-sampling in quasi-probability decompositions

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29	2-TUE	10:00 -11:30	713B	QALG	QALG::921::949::1255	Quantum Information & Miscellaneous	949	Sai Sakunthala Guddanti, Anil Prabhakar and Ria Rushin Joseph	Entropy Estimation in Multi-Qutrit Systems via Variational and Classical Neural Network Approaches
30	2-TUE	10:00 -11:30	713B	QALG	QALG::921::949::1255	Quantum Information & Miscellaneous	1255	Khrystyna Gnatenko	Entanglement Of Multi-Qubit Quantum Graph States And Studies Structural Properties Of Tripartite Graphs With Quantum Programming
31	2-TUE	13:00 -14:30	713B	QALG	QALG::236::372::948	Quantum Algorithms	236	Doron Gordon and Goren Gordon	Quantum Ray Marching: A CORDIC Implementation
32	2-TUE	13:00 -14:30	713B	QALG	QALG::236::372::948	Quantum Algorithms	372	Nishant Shukla	Coherent Rollout Oracles for Finite-Horizon Sequential Decision Problems
33	2-TUE	13:00 -14:30	713B	QALG	QALG::236::372::948	Quantum Algorithms	948	Simone Perriello, Alessandro Barenghi and Gerardo Pelosi	Improved Circuit Designs for Quantum Walk Search on Johnson Graphs
34	2-TUE	15:00 -16:30	713B	QALG	QALG::263::284::609	QAOA: Parameters & Initialization	263	Ankit Kulshrestha and Sarvagya Upadhyay	On the importance of hyperparameters in initializing parameterized quantum circuits
35	2-TUE	15:00 -16:30	713B	QALG	QALG::263::284::609	QAOA: Parameters & Initialization	284	Hyunwoo Kim and Youngseok Lee	SAFE ma-QAOA: Surrogate-Assisted and Fine-Tuning Enhanced Multi-Angle QAOA with Parameter Distillation
36	2-TUE	15:00 -16:30	713B	QALG	QALG::263::284::609	QAOA: Parameters & Initialization	609	Jose Falla, Bao Bach, Boris Tselikhovskiy and Ilya Saфро	Implications of Reduced DLAs for the Standard QAOA Ansatz
37	3-WED	10:00 -11:30	713B	QALG	QALG::157::589::1094	Quantum Error Correction: ML Decoders	157	Luca Menti and Francisco Lázaro	Neural Belief Matching Decoding for Topological Quantum Error Correction Codes
38	3-WED	10:00 -11:30	713B	QALG	QALG::157::589::1094	Quantum Error Correction: ML Decoders	589	Xiangjun Mi and Frank Mueller	Toward Uncertainty-Aware and Generalizable Neural Decoding for Quantum LDPC Codes
39	3-WED	10:00 -11:30	713B	QALG	QALG::157::589::1094	Quantum Error Correction: ML Decoders	1094	Sana Javed, Francisco Garcia-Herrero, Bane Vasić and Mark Flanagan	Belief Propagation-Informed Tesseract Decoder for Quantum LDPC Codes
40	3-WED	10:00 -11:30	705	QALG	QALG::1171::709::262	Quantum Optimization II	1171	Gabriel Thomaz, Eduarda Monteiro, Jerusa Marchi, Marcelo Pretto, Alisson Fumaco and Evandro Rosa	Second-Order FALQON Parameter Transfer for the Max-Cut Problem on 3-Regular Graphs
41	3-WED	10:00 -11:30	705	QALG	QALG::1171::709::262	Quantum Optimization II	709	Stuart Hadfield, Filip Maciejewski and Davide Venturelli	Noise-Directed Adaptive Remapping for Integer Optimization: from qubits to (encoded) qudits
42	3-WED	10:00 -11:30	705	QALG	QALG::1171::709::262	Quantum Optimization II	262	Triet Friedhoff, Mihir Metkar, Wade Davis, Vaibhaw Kumar and Alexey Galda	Pauli Correlation Encoding for mRNA Secondary Structure Prediction: Problem-Aware Decoding for Dense-Constraint QUBOs
43	3-WED	13:00 -14:30	713B	QALG	QALG::618::676::897	Error Detection & Mitigation I	618	Subrata Das, Archisman Ghosh and Swaroop Ghosh	A Graph-Based Forensic Framework for Inferring Hardware Noise of Cloud Quantum Backend
44	3-WED	13:00 -14:30	713B	QALG	QALG::618::676::897	Error Detection & Mitigation I	676	Sean Grzenda, Shahram Babaie and Chunming Qiao	Need One Bell-pair Only (NOBOL) for Low-Overhead Fault-Tolerant Quantum Computing
45	3-WED	13:00 -14:30	713B	QALG	QALG::618::676::897	Error Detection & Mitigation I	897	Yanis Le Fur, Ethan Egger, Hong-Ye Hu, Vincent Russo, William Zeng and Ryan LaRose	Opportunities and challenges in scaling quantum error detection on hardware
46	3-WED	15:00 -16:30	713B	QALG	QALG::361::366::756	Circuit Optimization & Synthesis	361	Till Schnittka, Majd Assaad, Abhoy Kole and Rolf Drechsler	Low T-Depth Clifford+T Decomposition of Large Toffoli Gates using Dynamic Phase Correction
47	3-WED	15:00 -16:30	713B	QALG	QALG::361::366::756	Circuit Optimization & Synthesis	366	Antoine Lemelin, Olivier Landon-Cardinal, Christophe Pere and Camille Coti	Mid-circuit measurement as an algorithmic primitive for variational algorithms
48	3-WED	15:00 -16:30	713B	QALG	QALG::361::366::756	Circuit Optimization & Synthesis	756	Chih-Hsiang Chuang, Tian-Fu Chen, Jie-Hong Roland Jiang, Bo-Hung Chen and Dah-Wei Chiou	Boolean Matching for Quantum Circuits
49	4-THU	10:00 -11:30	713B	QALG	QALG::306::317::888	Quantum Performance & Complexity	306	Nicholas Sherman, Hrant Gharibyan, Zuhair Mullath, Vincent Su, Hayk Tepanyan and Yuxuan Zhang	Heuristic Quantum Advantage with Peaked Circuits
50	4-THU	10:00 -11:30	713B	QALG	QALG::306::317::888	Quantum Performance & Complexity	317	Leonardo Placidi, Enrico Rinaldi, Keisuke Fujii and Chen-Yu Liu	The Impact of Qubit Connectivity on Quantum Advantage in Noisy IQP Circuits
51	4-THU	10:00 -11:30	713B	QALG	QALG::306::317::888	Quantum Performance & Complexity	888	Fei Meng, Bin Cheng, Jianan Li and Man-Hong Yung	Weak Permanent Anti-Concentration for Random Gaussian Matrices in Boson Sampling
52	4-THU	13:00 -14:30	713B	QALG	QALG::1187::561::426	Quantum Optimization I	1187	Phillip Lotshaw, Titus Morris, Stuart Hadfield and Ryan Bennink	Iterative warm-start optimization with quantum imaginary time evolution
53	4-THU	13:00 -14:30	713B	QALG	QALG::1187::561::426	Quantum Optimization I	561	Cameron Ibrahim, Bao Bach, Jad Salem, Reuben Tate, Kien Nguyen, Stephan Eidenbenz and Ilya Saфро	Quantum Hypergraph Partitioning
54	4-THU	13:00 -14:30	713B	QALG	QALG::1187::561::426	Quantum Optimization I	426	Shunta Ide and Shu Tanaka	Unfair Sampling of Quantum Annealing in Weighted Graph Bipartitioning Problems
55	4-THU	15:00 -16:30	713B	QALG	QALG::360::747::1164	Quantum Arithmetic & Reversible Circuits	360	Lei-Han Yao, Shang-Wei Lin, Yu-Chung Chen and Yean-Ru Chen	A Modular and T-Gate Efficient Architecture for Quantum Leading-Zero/One Counter
56	4-THU	15:00 -16:30	713B	QALG	QALG::360::747::1164	Quantum Arithmetic & Reversible Circuits	747	Iain Burge, Michel Barbeau and Joaquin Garcia-Alfaro	Quantum CORDIC — Arcsine on a Budget
57	4-THU	15:00 -16:30	713B	QALG	QALG::360::747::1164	QAOA for Optimization Problems	1164	Lucas Braydwood and Phillip Lotshaw	QAOA Parameter Transfer for Hypergraphs

QSYS - Quantum Systems Software

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v123	Day	Time	Room	Track	QCE-PRG-ID	Session Name	ECID	Paper Authors	Paper Title
1	0-SUN	10:00 -11:30	801A	QSYS	QSYS::428::604::905	Quantum Circuit Compilation Under Constraints	428	Justin Kalloor, Mathias Weiden, Ed Younis, John Kubiawicz and Costin Iancu	Multi-Qubit Dyadic Phase Fixing for Fault-Tolerant Quantum Compilation
2	0-SUN	10:00 -11:30	801A	QSYS	QSYS::428::604::905	Quantum Circuit Compilation Under Constraints	604	Abhishek Shringi, Ahmed Shokry, Movahhed Sadeghi and Mahmut Kandemir	QLego : A Cross-SDK Framework for Optimal Quantum Compilation
3	0-SUN	10:00 -11:30	801A	QSYS	QSYS::428::604::905	Quantum Circuit Compilation Under Constraints	905	Jack Weinberg and Narayanan Rengaswamy	Stabilizers for Compiling Logical Circuits under Hardware Constraints
4	0-SUN	10:00 -11:30	801B	QSYS	QSYS::215::427::662	Runtime Workflows & Compilation Reuse	215	Mar Tejedor, Javier Conejero and Rosa M. Badia	A Semantic Quantum Circuit Cache for Scalable and Distributed Quantum-Classical Workflows
5	0-SUN	10:00 -11:30	801B	QSYS	QSYS::215::427::662	Runtime Workflows & Compilation Reuse	427	Zhixin Song, Aniket Dalvi, Nils Quetschlich, Yi-Ting Chen and Ryan Shaffer	Improving Amazon Braket Workloads through Application-Aware Quantum Compilation
6	0-SUN	10:00 -11:30	801B	QSYS	QSYS::215::427::662	Runtime Workflows & Compilation Reuse	662	Muhammad Arslan Ansari, Salvatore Zammuto and Martin Schulz	Re-using Runtime Quantum Circuit Transpilation in ADAPT-VQE Workflows
7	0-SUN	13:00 -14:30	801A	QSYS	QSYS::267::307::333	AI-Assisted Quantum Software Engineering	267	Qiming Li and Jianjun Zhao	Q-AssertBench: Benchmarking Large Language Models on Assertion Generation in Quantum Programs
8	0-SUN	13:00 -14:30	801A	QSYS	QSYS::267::307::333	AI-Assisted Quantum Software Engineering	307	Pietro Cassieri, Giuseppe Scanniello, Seung Yeob Shin, Fabrizio Pastore and Domenico Bianculli	Beyond Rules: LLM Powered Linting for Quantum Programs
9	0-SUN	13:00 -14:30	801A	QSYS	QSYS::267::307::333	AI-Assisted Quantum Software Engineering	333	Andres Paz, Sarju Patel, Cordelia Li, Sylvie Lausier, Christian Tarta and Mayee Sun	StabilizerBench: A Benchmark for AI-Assisted Quantum Error Correction Circuit Synthesis
10	0-SUN	13:00 -14:30	801B	QSYS	QSYS::252::957::1122	Quantum Software Practice & Observability	252	Otso Kinanen, Valter Uotila and Vlad Stirbu	Using Experiment Tracking to Enhance Quantum Software Development Process: A Case Study of Quantum Reservoir Computing with Error Mitigation
11	0-SUN	13:00 -14:30	801B	QSYS	QSYS::252::957::1122	Quantum Software Practice & Observability	957	Ronja Heikkinen, Santiago Núñez-Corrales and Vlad Stirbu	Navigating the Socio-Technical Complexity Challenge in Quantum Software Ecosystems
12	0-SUN	13:00 -14:30	801B	QSYS	QSYS::252::957::1122	Quantum Software Practice & Observability	1122	Yuto Morohashi, Hitomi Takahashi, Naoki Kanazawa, Yukio Kawashima, Hiroshi Horii and Kengo Nakajima	Observability Architecture for Quantum-Centric Supercomputing Workflows
13	0-SUN	15:00 -16:30	801A	QSYS	QSYS::280::738::1123	Quantum Program Testing & Denoising	280	Maryse Ernzer, Seung Yeob Shin, Fabrizio Pastore and Domenico Bianculli	Randomized and Diverse Input State Generation for Quantum Program Testing
14	0-SUN	15:00 -16:30	801A	QSYS	QSYS::280::738::1123	Quantum Program Testing & Denoising	738	Masaomi Yamaguchi and Fuyuki Ishikawa	Symmetry-Oriented Testing Framework for Variational Quantum Algorithms
15	0-SUN	15:00 -16:30	801A	QSYS	QSYS::280::738::1123	Quantum Program Testing & Denoising	1123	Ning Ma, Jun Dai and Heng Li	Backend-Aware Graph Learning for Denoising Outcome Distributions in Quantum Program Testing
16	0-SUN	15:00 -16:30	801B	QSYS	QSYS::343::411::912	Quantum System Characterization	343	Yi-Ting Chen, Aniket Dalvi and Nils Quetschlich	Resource-Optimal Screening for Two-Tier Quantum System Characterization
17	0-SUN	15:00 -16:30	801B	QSYS	QSYS::343::411::912	Quantum System Characterization	411	Cenk Tüysüz, Abhijith Jayakumar, Carleton Coffrin, Marc Vuffray and Andrey Lokhov	Learning response functions of analog quantum computers: analysis of neutral-atom and superconducting platforms
18	0-SUN	15:00 -16:30	801B	QSYS	QSYS::343::411::912	Quantum System Characterization	912	Vincent Mutolo, Devon Campbell, Quinn Manning, Henri Dubourg, Ruibin Lyu, Simha Sethumadhavan, Dan Rubenstein and Salvatore Stolfo	Quantum Computer Fingerprinting using Error Syndromes
19	1-MON	10:00 -11:30	717A	QSYS	QSYS::41::218:1157	Security, Trust, & Attacks	41	Theodoros Trochatos, Chuanqi Xu, Sanjay Deshpande, Yao Lu, Yongshan Ding and Jakub Szefer	SoteriaQ: Secure Quantum Computation via Gate-Level Obfuscation from Untrusted Clouds
20	1-MON	10:00 -11:30	717A	QSYS	QSYS::41::218:1157	Security, Trust, & Attacks	218	Anthony Etim and Jakub Szefer	Fault Attacks on ML-based Quantum Control and Error Correction
21	1-MON	10:00 -11:30	717A	QSYS	QSYS::41::218:1157	Security, Trust, & Attacks	1157	Shashvat Shukla, Dan Browne and Shin Nishio	Anticipating decoder side-channel attacks in fault-tolerant quantum computers
22	1-MON	10:00 -11:30	803A	QSYS	QSYS::585::874::971	Learning-Based Circuit Synthesis	585	Mehdi Saeedi, Eddie Richter and Paul Hartke	When Close Enough Is Not Enough: Autoregressive Drift in Quantum Circuit Synthesis
23	1-MON	10:00 -11:30	803A	QSYS	QSYS::585::874::971	Learning-Based Circuit Synthesis	874	Daniele Lizzio Bosco, Jacopo Cossio, Carla Piazza and Giuseppe Serra	AlphaClifford: Efficient Clifford Synthesis and Transpilation with Model-based RL
24	1-MON	10:00 -11:30	803A	QSYS	QSYS::585::874::971	Learning-Based Circuit Synthesis	971	Christian Rasmussen, Mohammad Walid Charrwi, Ed Younis, Wibe Albert De Jong and Samah Saeed	Breaking the Locality Barrier in Approximate Quantum Synthesis with Reinforcement Learning
25	1-MON	10:00 -11:30	717B	QSYS	QSYS::566::603::648	Noise Learning & Runtime Calibration	566	Maxwell Poster, Jason Chadwick and Jonathan Baker	EDIQT: Enabling Dynamism for In-Situ Quantum Calibration Technologies
26	1-MON	10:00 -11:30	717B	QSYS	QSYS::566::603::648	Noise Learning & Runtime Calibration	603	Ryo Sakai and Yu Yamashiro	Quantum hardware noise learning via differentiable Kraus representation on tensor networks

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27	1-MON	10:00 -11:30	717B	QSYS	QSYS::566::603::648	Noise Learning & Runtime Calibration	648	Xiaolong Deng	Runtime Calibration as State-Trajectory Feedback Control in Quantum-Classical Workflows
28	1-MON	13:00 -14:30	803A	QSYS	QSYS::796::879::881	Gate, Oracle, & Unitary Synthesis	796	Evan McKinney and Lev Bishop	GULPS: Two-Qubit Gate Synthesis via Linear Programming for Heterogeneous Instruction Sets
29	1-MON	13:00 -14:30	803A	QSYS	QSYS::796::879::881	Gate, Oracle, & Unitary Synthesis	879	Frans Perkkola, Arianne Meijer-van de Griend and Jukka K. Nurminen	Architecture-aware Unitary Synthesis
30	1-MON	13:00 -14:30	803A	QSYS	QSYS::796::879::881	Gate, Oracle, & Unitary Synthesis	881	David Clarino, Chitranshu Arya, Zhanhe Qi and Shigeru Yamashita	Improving Phase Oracle Synthesis using Logic Network Based Synthesis and ESOP Minimization
31	1-MON	15:00 -16:30	803A	QSYS	QSYS::873::990::1057	Pulse-Level Control & Quantum Sensing	873	Masoud Hakimi Heris, Yuan Liu and Frank Mueller	HyPulse: A Pulse Synthesis Framework for Hybrid Qubit-Oscillator Gates on Trapped-Ion Platform
32	1-MON	15:00 -16:30	803A	QSYS	QSYS::873::990::1057	Pulse-Level Control & Quantum Sensing	990	Gabrielle MacNeil and Qiaoyan Yu	Q-STEP: A Cross-Layer Design and Validation Framework for Advancing Quantum Sensing
33	1-MON	15:00 -16:30	803A	QSYS	QSYS::873::990::1057	Pulse-Level Control & Quantum Sensing	1057	Anthony Santana, Alex McCaskey, Shane Caldwell and Krysta Svore	A Linearly Typed MLIR Dialect for Pulse-Level Quantum Control Scheduling
34	3-WED	10:00 -11:30	713A	QSYS	QSYS::279::511::845	Scalable Quantum Circuit Simulation	279	Satoshi Imamura, Haruto Fujii, Nozomi Aoki, Akihiro Tabuchi, Ryuta Kawano and Akihiko Kasagi	QompSim: A Distributed State-Vector Simulator with Lossy and All-Zero Compression
35	3-WED	10:00 -11:30	713A	QSYS	QSYS::279::511::845	Scalable Quantum Circuit Simulation	511	Adeeb Kabir, Steven Nguyen, Sohan Ghosh, James Keppens, Tijl Kiran, Yipeng Huang, Isaac Kim and Bart Sorée	Sdim: A High-Dimensional Stabilizer Simulator
36	3-WED	10:00 -11:30	713A	QSYS	QSYS::279::511::845	Scalable Quantum Circuit Simulation	845	Tim Littau and Rihan Hai	QuWARP: A Workload-Aware Reuse Planner for simulating Quantum Circuits
37	3-WED	13:00 -14:30	713A	QSYS	QSYS::341::395::1076	Real-Time QEC Decoding	341	Zhengyu Cai, Mohamad Ali Jarkas, Bokun Zhao and Warren Gross	Dithered Belief Propagation for Real-Time Quantum Memory Decoding
38	3-WED	13:00 -14:30	713A	QSYS	QSYS::341::395::1076	Real-Time QEC Decoding	395	Ameya S. Bhave, Navnil Choudhury and Kanad Basu	DART-Q : A Deadline-Driven Framework for Real-Time QLDPC Decoding
39	3-WED	13:00 -14:30	713A	QSYS	QSYS::341::395::1076	Real-Time QEC Decoding	1076	Daniel Bascones, Arshpreet Singh Maan, Valentin Savin and Francisco García-Herrero	A Scalable FPGA Architecture for Real-Time Decoding of Quantum LDPC Codes Using GARI
40	3-WED	15:00 -16:30	713A	QSYS	QSYS::296::790::791	Modular Architectures & Communication-Aware Compilation	296	Enrico Russo, Francesco Giulio Blanco, Elio Vinciguerra, Davide Patti, Giuseppe Ascia and Maurizio Palesi	COSMA: Communication-aware Optimization of Fermionic Simulation Kernels for Modular Quantum Architectures
41	3-WED	15:00 -16:30	713A	QSYS	QSYS::296::790::791	Modular Architectures & Communication-Aware Compilation	790	Mingyoung Jessica Jeng, Nikola Vuk Maruszewski, Mu-Tse Lau, Connor Selna, Michael Gavrincea, Kaitlin N. Smith and Nikos Hardavallac	SEQC: Stratify-Elaborate Quantum Compilation Towards Modular Hybrid Architectures
42	3-WED	15:00 -16:30	713A	QSYS	QSYS::296::790::791	Modular Architectures & Communication-Aware Compilation	791	Vinoth Rao Kulkarni, Jaehyun Lee, Lauren Li, Aaron Orenstein, Xinpeng Li, Shuai Xu, Daniel Blankenberg and Vipin Chaudhary	InterQ: Communication-Aware Scheduling Across Modular QPUs with Classical and Quantum Links
43	4-THU	10:00 -11:30	713A	QSYS	QSYS::49::575::840	Accelerated QEC & Feedback Systems	49	Alessio Cicero, Luigi Altamura, Moritz Lange, Mats Granath and Pedro Trancoso	Low Latency GNN Accelerator for Quantum Error Correction
44	4-THU	10:00 -11:30	713A	QSYS	QSYS::49::575::840	Accelerated QEC & Feedback Systems	575	Tina Oberoi, Joshua Vizslai and Frederic T. Chong	ADaPT: Adaptive-window Decoding for Practical fault-Tolerance
45	4-THU	10:00 -11:30	713A	QSYS	QSYS::49::575::840	Accelerated QEC & Feedback Systems	840	Junyi Liu, Yi Lee, Yilun Xu, Gang Huang and Xiaodi Wu	A Scalable Open-Source QEC System with Sub-Microsecond Decoding-Feedback Latency
46	4-THU	13:00 -14:30	713A	QSYS	QSYS::382::668::830	QLDPC & Error-Corrected Architectures	382	Kun Liu, Takahiro Tsunoda, Sophia H. Xue, Evan McKinney, Zeyuan Zhou, Shifan Xu, Robert J Schoelkopf and Yongshan Ding	Efficient Routing of Quantum LDPC Codes on Programmable 2D Toric Architectures
47	4-THU	13:00 -14:30	713A	QSYS	QSYS::382::668::830	QLDPC & Error-Corrected Architectures	668	Jason D. Chadwick and Frederic T. Chong	CABLECAR: efficiently scheduling QLDPC codes on a tileable spin qubit chip with shuttling
48	4-THU	13:00 -14:30	713A	QSYS	QSYS::382::668::830	QLDPC & Error-Corrected Architectures	830	Julian Shen, Ludwig Schmid and Robert Wille	Routing Techniques for Error Corrected Silicon Spin Qubit Quantum Architectures
49	4-THU	15:00 -16:30	713A	QSYS	QSYS::243::770::934	Distributed Quantum Systems	243	Evan Sutcliffe and Coral Westoby	Tolerating Device Failure in Distributed Quantum Computing
50	4-THU	15:00 -16:30	713A	QSYS	QSYS::243::770::934	Distributed Quantum Systems	770	Afonso Azenha, Ilia Polian and Sebastian Brandhofer	Scalable Transpilation for Overcoming Restricted Connectivity in Distributed Superconducting Quantum Architectures
51	4-THU	15:00 -16:30	713A	QSYS	QSYS::243::770::934	Distributed Quantum Systems	934	Joost Van Veen, Luise Prielinger and Sebastian Feld	Rethinking How to Act: Action-Space Engineering for Reinforcement Learning-Based Circuit Routing in Distributed Quantum Systems
52	5-FRI	10:00 -11:30	803A	QSYS	QSYS::211::521::701	Quantum Languages, Semantics, & Model Checking	211	Qiming Li, Shangzhou Xia and Jianjun Zhao	Abstraction-Driven State-Space Reduction for Quantum Model Checking
53	5-FRI	10:00 -11:30	803A	QSYS	QSYS::211::521::701	Quantum Languages, Semantics, & Model Checking	521	David Chamizo, Jose Garcia-Alonso and Juan M. Murillo	High-level quantum structured programs as quantum registers compositions
54	5-FRI	10:00 -11:30	803A	QSYS	QSYS::211::521::701	Quantum Languages, Semantics, & Model Checking	701	Ulrik de Muelenaere, Amr Sabry and Peter M. Kogge	Correcting Misconceptions About Quantum-Controlled Measurement

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55	5-FRI	10:00 -11:30	701B	QSYS	QSYS::406::736::1087	Qubit Mapping & Layout Selection	406	Brent Russon, Bao Bach, Ed Younis and Ilya Safro	Scaling Qubit Mapping and Routing With Position Graph Abstraction and Memoization
56	5-FRI	10:00 -11:30	701B	QSYS	QSYS::406::736::1087	Qubit Mapping & Layout Selection	736	Matthias Reumann, Yannick Stade, Robert Wille and Lukas Burgholzer	MLIR for Quantum Beyond Gate Cancellation: Quantum Circuit Mapping Reimagined
57	5-FRI	10:00 -11:30	701B	QSYS	QSYS::406::736::1087	Qubit Mapping & Layout Selection	1087	Yulun Wang, Esteban Ginez, Jamie Friel, Yuval Baum, Jin-Sung Kim, Alex Shih and Oded Green	Δ -Motif: Parallel Subgraph Isomorphism via Tabular Operations for Scalable Layout Selection
58	5-FRI	10:00 -11:30	713B	QSYS	QSYS::613::870::1093	Evaluation Metrics & Comparison	613	Siddarth Shinde and Jakub Szefer	Quantum Fidelity-per-Cost: A Metric for Evaluation of Quantum Computing Systems
59	5-FRI	10:00 -11:30	713B	QSYS	QSYS::613::870::1093	Evaluation Metrics & Comparison	870	Dominik Köster and Wolfgang Mauerer	Claim against Measurement: Statistical Artefacts in Quantum Error Mitigation Benchmarks
60	5-FRI	10:00 -11:30	713B	QSYS	QSYS::613::870::1093	Evaluation Metrics & Comparison	1093	Emil Khusainov, Yanbin Chen, Jonas Winklmann, Helmut Seidl and Christian Mendl	Practical Insights into Fair Comparison and Evaluation Frame for Neutral-Atom Compilers
61	5-FRI	10:00 -11:30	701A	QSYS	QSYS::564::641::654	Quantum Systems & Resource Management	564	Charles Cao, Sergei Kalinin, Jong Youl Choi and Ahmad Maroof Karimi	TRACE: Temporal Residency-AwareCircuit Execution for Over-Capacity Quantum Hardware
62	5-FRI	10:00 -11:30	701A	QSYS	QSYS::564::641::654	Quantum Systems & Resource Management	641	Archisman Ghosh, Avimita Chatterjee and Swaroop Ghosh	No Tile Left Behind: Multiprogramming for Surface-Code Architectures
63	5-FRI	10:00 -11:30	701A	QSYS	QSYS::564::641::654	Quantum Systems & Resource Management	654	Srikar Chundury, Seongmin Kim, Muralikrishnan Gopalakrishnan Meena, Chao Lu, In-Saeng Suh and Frank Mueller	SeleQtor: Intelligent Quantum Backend Selection and Runtime Estimation
64	5-FRI	13:00 -14:30	701B	QSYS	QSYS::234::240::459	Quantum Circuit Optimization & Compiler Diagnostics	234	Finn Voichick, Leonidas Lampropoulos and Robert Rand	Circuit Optimization via Gradients and Noise-Aware Compilation
65	5-FRI	13:00 -14:30	701B	QSYS	QSYS::234::240::459	Quantum Circuit Optimization & Compiler Diagnostics	240	Xiao-Ting Michelle To, Nils Quetschlich, Amr Elsharkawy, Martin Schulz, Robert Wille and Dieter Kranzlmüller	Evaluating the Effect of the Order of Optimization Passes in Quantum Circuit Optimization
66	5-FRI	13:00 -14:30	701B	QSYS	QSYS::234::240::459	Quantum Circuit Optimization & Compiler Diagnostics	459	Chandrachud Pati and Yogesh Simmhan	Per-Phase Fidelity Attribution for Quantum Compilers using HBR Decomposition
67	5-FRI	13:00 -14:30	713B	QSYS	QSYS::345::607::1029	Benchmarking & Reproducibility	345	Zachary Macaskill-Smith, Unmol Sharma, Melissa Warner, Kalman Varga and David Hyde	Ember: An Extensible Benchmark Suite for Quantum Annealing Embedding Algorithms
68	5-FRI	13:00 -14:30	713B	QSYS	QSYS::345::607::1029	Benchmarking & Reproducibility	607	Haoyuan Tian and Jie Ren	Beyond Qubit Count in Practice: End-to-End Characterization of Quantum Circuit Simulation Costs
69	5-FRI	13:00 -14:30	713B	QSYS	QSYS::345::607::1029	Benchmarking & Reproducibility	1029	Dominik Köster, Maja Franz, Benjamin Zec, Nicole Hoess, Ralf Ramsauer and Wolfgang Mauerer	Works on My QPU: Reproducibility in Quantum Computing Research
70	5-FRI	13:00 -14:30	803A	QSYS	QSYS::469::571::684	Surface-Code & Lattice-Surgery Scheduling	469	Shintaro Sato and Yasunari Suzuki	Scheduling of syndrome measurements with a few ancillary qubits
71	5-FRI	13:00 -14:30	803A	QSYS	QSYS::469::571::684	Surface-Code & Lattice-Surgery Scheduling	571	Lakshika Rathi, Pau Escofet, Joshua Vizslai and Margaret Martonosi	X-Z Round Scheduling for the Surface Code with Defects under Biased Noise
72	5-FRI	13:00 -14:30	803A	QSYS	QSYS::469::571::684	Surface-Code & Lattice-Surgery Scheduling	684	Steven Hofmeyr, Mathias Weiden, Justin Kalloor, John D. Kubitowicz and Costin Iancu	PureMagic: A Dynamic Scheduler for Lattice Surgery
73	5-FRI	13:00 -14:30	701A	QSYS	QSYS::530::933::1074	Foundations & Algorithms for Quantum Computing	530	Yannik Böck, Holger Boche and Frank Fitzek	Feynman Meets Turing: The Curse of Quantum Universality
74	5-FRI	13:00 -14:30	701A	QSYS	QSYS::530::933::1074	Foundations & Algorithms for Quantum Computing	933	Oxana Shaya, Konstantin Golovkin, Mainak Roy and Vincent Russo	PauLie: Fast Classification of Pauli Dynamical Lie Algebras
75	5-FRI	13:00 -14:30	701A	QSYS	QSYS::530::933::1074	Foundations & Algorithms for Quantum Computing	1074	Dhanvi Bharadwaj, Yuewen Hou, Guangyi Li and Gokul Subramanian Ravi	Scalable Clifford-Based Classical Initialization for the Quantum Approximate Optimization Algorithm
76	5-FRI	15:00 -16:30	701B	QSYS	QSYS::247::1203::125	Quantum Circuit Cutting & Partitioned Execution	247	Fiona Jiali Fröhler, Yannick Stade, Christian Ufrecht, Daniel D. Scherer and Robert Wille	Scalable Circuit Cutting: A Framework for Combined Gate and Wire Cuts Using Gate Groups
77	5-FRI	15:00 -16:30	701B	QSYS	QSYS::247::1203::125	Quantum Circuit Cutting & Partitioned Execution	1203	Qingwen Liu, Xinpeng Li, Ji Liu, Xiaokun Yang, Lauren Li, Yifan Yu, Yang Zhang, Liran Ma, Shuai Xu and Vipin Chaudhary	Non-Separable Wire Cutting: Enhancing Quantum Compilation
78	5-FRI	15:00 -16:30	701B	QSYS	QSYS::247::1203::125	Quantum Circuit Cutting & Partitioned Execution	1254	Frej Larssen, Luca Pennati, Erik M. Åsgrim, Ivy Peng and Stefano Markidis	Fewer Histories, Faster Patns: Distributed Quantum Circuit Feynman Simulation via History Reduction, Checkpointing, and Pruning
79	5-FRI	15:00 -16:30	713B	QSYS	QSYS::441::527::576	Quantum Control Architectures	441	Xiaorang Guo, Jonas Winklmann, Vengkeat Chea and Martin Schulz	AtomFlow: An End-to-End FPGA-Based Control Architecture for Neutral Atom Quantum Computers
80	5-FRI	15:00 -16:30	713B	QSYS	QSYS::441::527::576	Quantum Control Architectures	527	Luca Otting, Xiaorang Guo, Emmanouil Giortamis, Benjamin Lienhard, Pramod Bhatotia and Martin Schulz	Multi-Stage Mamba-Based Architecture for Fast and Scalable Superconducting Qubit Readout
81	5-FRI	15:00 -16:30	713B	QSYS	QSYS::441::527::576	Quantum Control Architectures	576	Xiaorang Guo, Kun Qin, Yanbin Chen, Carsten Trinitis and Martin Schulz	Vectorizing Quantum Control: A RISC-V Vector Extension Architecture for Scalable Qubit Systems
82	5-FRI	15:00 -16:30	701A	QSYS	QSYS::464::574::980	Execution in Fault-Tolerant Quantum Computing	464	Boshuai Ye, Arif Ali Khan and Peng Liang	When T-Depth Misleads: Predicting Fault-Tolerant Quantum Execution Slowdown under Magic-State Delivery Constraints

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83	5-FRI	15:00 -16:30	701A	QSYS	QSYS::464::574::980	Execution in Fault-Tolerant Quantum Computing	574	Aditi Awasthi, Sayam Sethi, Sahil Khan, Gokul Ravi and Jonathan Baker	Price and Payoff: Non-Determinism in Fault Tolerant Quantum Computation
84	5-FRI	15:00 -16:30	701A	QSYS	QSYS::464::574::980	Execution in Fault-Tolerant Quantum Computing	980	Aditi Awasthi, Gokul Ravi and Jonathan Baker	Stalls and Spequlation: Pipelined Execution for Fault Tolerant Quantum Computation
85	5-FRI	15:00 -16:30	803A	QSYS	QSYS::707::820::1249	Fault-Tolerant Protocols & Magic-State Generation	707	Asif Ronggon and Tasnuva Farheen	Adaptive Error Budget Allocation for Fault-Tolerant Quantum Resource Estimation: A Metaheuristic Approach
86	5-FRI	15:00 -16:30	803A	QSYS	QSYS::707::820::1249	Fault-Tolerant Protocols & Magic-State Generation	820	Sayam Sethi, Sahil Khan, Aditi Awasthi, Abhinav Anand and Jonathan Baker	INJEQT: Improved Magic-State Injection Protocol for Fault-Tolerant Quantum Extractor Architectures
87	5-FRI	15:00 -16:30	803A	QSYS	QSYS::707::820::1249	Fault-Tolerant Protocols & Magic-State Generation	1249	Mohammad Rowshan	Structural Conditions for Native CCZ Magic-State Fountains in qLDPC Codes

QNET - Quantum Networking Communications

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v123	Day	Time	703	Track	QCE-PRG-ID	Session Name	ECID	Paper Authors	Paper Title
1	0-SUN	10:00 -11:30	705	QNET	QNET::335::499	Real-Time Control & Benchmarking for Quantum Networks	335	You-Wei Cheah, Swapnil Patel, Elia Perego, Se-young Yu, Justin Philips, Guangqi Gao, Erhan Saglamyurek, Wenji Wu, Hartmut Häeffner and Inder Monga	Real-time control of trapped-ion quantum networks
2	0-SUN	10:00 -11:30	705	QNET	QNET::335::499	Real-Time Control & Benchmarking for Quantum Networks	499	Samuel Oslovich, Stephanie Wehner and Claudio Cicone	A Full-System Benchmarking Suite for Quantum Networks
3	0-SUN	13:00 -14:30	705	QNET	QNET::250::649::1183	Physical-Layer Optical Links & Stabilization	250	Fabian Klingmann, Joscha Hanel, Raul Lahoz Sanz, Vincent Rehlinger, Mira Stephan, René Kirrbach, Alexander Noack, Jingzhong Yang and Fei Ding	Comparison of Polarization Compensation Methods for Quantum Key Distribution
4	0-SUN	13:00 -14:30	705	QNET	QNET::250::649::1183	Physical-Layer Optical Links & Stabilization	649	Abderrahim Amlou, Amar Abane, Cory Nunn, Jabir Marakkarakath Vadakkepurayil, Abdella Battou, Van Sy Mai and Ahmed Lbath	Physics-Informed Discrete-Event Simulation of Polarization-Encoded Quantum Networks
5	0-SUN	13:00 -14:30	705	QNET	QNET::250::649::1183	Physical-Layer Optical Links & Stabilization	1183	Laura D'Avossa, Angela Sara Cacciapuoti and Marcello Caleffi	Interconnection of Quantum Networks at Urban scale: Analysis of Temporal Stability of Entangled Photon Sources
6	0-SUN	15:00 -16:30	705	QNET	QNET::129::261::699	Entanglement Distillation & Purification Scheduling	129	Siddharth Chander, Xinan Chen and Allen Zang	Entanglement Distillation and Swapping Scheduling in Quantum Repeaters with Noisy Memories
7	0-SUN	15:00 -16:30	705	QNET	QNET::129::261::699	Entanglement Distillation & Purification Scheduling	261	Marcel Kokorsch and Guido Dietl	Optimal Entanglement Distillation Scheduling in Repeater-Based QKD Beyond Fidelity Constraints
8	0-SUN	15:00 -16:30	705	QNET	QNET::129::261::699	Entanglement Distillation & Purification Scheduling	699	Anoosha Fayyaz, Prashant Krishnamurthy, Kaushik Seshadreesan, David Tipper and Amy Babay	To Purify or Not to Purify: Entanglement Purification under Input Fidelity Asymmetry in Quantum Networks
9	1-MON	10:00 -11:30	705	QNET	QNET::419::704::1198	Teleportation & Entanglement-Fidelity Engineering	419	Mahmoud Saad Abouamer, Jakob Kaltoft Søndergaard and Petar Popovski	Latency-Constrained Encoded Quantum Teleportation with Punctured Codes
10	1-MON	10:00 -11:30	705	QNET	QNET::419::704::1198	Teleportation & Entanglement-Fidelity Engineering	704	Mohadeseh Azari, Anoosha Fayyaz, Amy Babay, David Tipper, Prashant Krishnamurthy and Kaushik Seshadreesan	Toward Hop-Independent Fidelity in Quantum Data Centers: Resource Requirements for Entanglement Purification
11	1-MON	10:00 -11:30	705	QNET	QNET::419::704::1198	Teleportation & Entanglement-Fidelity Engineering	1198	Lia Suci Waliani, Georges Kaddoum, Mahdi Chehimi and Shahan Hawatian	Distillation-Enhanced Continuous-Variable Quantum Teleportation for Satellite Communication Networks
12	1-MON	13:00 -14:30	705	QNET	QNET::352::1167::1135	Entanglement Routing & Network Scheduling	352	Dillon Horton, Hien Hoang, Ethan Hunt, Dheeraj Tukkugudam, Tu Nguyen and Kun Suo	Towards Connected k-Entanglement Routing Paths in Quantum Networks
13	1-MON	13:00 -14:30	705	QNET	QNET::352::1167::1135	Entanglement Routing & Network Scheduling	1167	Sadgun Ram Dasi and Dr.Aswath Babu H	Priority-Aware Routing for Quantum Networks: Integrating Coherence-Time Constraints into Scheduling
14	1-MON	13:00 -14:30	705	QNET	QNET::352::1167::1135	Entanglement Routing & Network Scheduling	1135	Maria Caterina D'Aloia and Christian Esposito	Predictive Cluster-Aware Recursive Routing for QKD-FL Networks
15	1-MON	15:00 -16:30	703	QNET	QNET::638::946::1224	QKD Network Planning in Terrestrial & Non-Terrestrial Networks	638	Tianqu Luo, Ibrahim Arslantas, Sezin Ozturk, Qiaolun Zhang, Alberto Sebastián-Lombraña, Mehdi Bolourian, Jiaheng Xiong, Francesco Musumeci, Vicente Martín, Raouf Boutaba and Massimo Tornatore	Network Planning with Trusted-Relay Minimization for Measurement-Device-Independent QKD Networks
16	1-MON	15:00 -16:30	703	QNET	QNET::638::946::1224	QKD Network Planning in Terrestrial & Non-Terrestrial Networks	946	Anjum Ashraaf, Syed Shah Mir, Mohamed Abdallah, Ala Al-Fuqaha, Aiman Erbad and Mounir Hamdi	Trust-Aware Joint Scheduling and Key Routing for UAV-Assisted TF-QKD Networks
17	1-MON	15:00 -16:30	703	QNET	QNET::638::946::1224	QKD Network Planning in Terrestrial & Non-Terrestrial Networks	1224	Rahi Shah and Di Wu	A System-Level Feasibility Framework for Key-Constrained Secure Communication in Cislunar Quantum Networks
18	2-TUE	10:00 -11:30	604	QNET	QNET::255::1218::1267	Quantum Network Tomography & Entanglement Certification	255	Xuchuang Wang, Joseph C Chapman, Matheus Guedes de Andrade, Yu-Zhen Janice Chen, Joseph M. Lukens, Gayane Vardoyan and Don Towsley	Quantum-Classical Coexisting Network Tomography
19	2-TUE	10:00 -11:30	604	QNET	QNET::255::1218::1267	Quantum Network Tomography & Entanglement Certification	1218	Yufei Zheng, Zihao Gong, Saikat Guha and Don Towsley	Quantum-enhanced Network Tomography
20	2-TUE	10:00 -11:30	604	QNET	QNET::255::1218::1267	Quantum Network Tomography & Entanglement Certification	1267	Xuan Du Trinh, Zhengyu Wu, Junlin Bai, Huan-Hsin Tseng, Nengkun Yu and Aruna Balasubramanian	Entanglement certification by measuring nonlocality
21	2-TUE	13:00 -14:30	604	QNET	QNET::757::783::1146	QKD Protocol Security & Post-Processing	757	David Polzoni, Tommaso Bianchi and Mauro Conti	Strengthening security and noise resistance in one-way quantum key distribution protocols through hypercube-based quantum walks
22	2-TUE	13:00 -14:30	604	QNET	QNET::757::783::1146	QKD Protocol Security & Post-Processing	783	Walter Krawec	Finite Key Security of the Extended B92 Protocol
23	2-TUE	13:00 -14:30	604	QNET	QNET::757::783::1146	QKD Protocol Security & Post-Processing	1146	Saul Hafting, Darshana Upadhyay, Sagarika Ghosh and Srinivas Sampalli	QRNG-Enhanced Quantum Key Distribution with Adaptive Authentication and Key Derivation
24	2-TUE	15:00 -16:30	604	QNET	QNET::650::708::710	Distributed Quantum Computing & Error Correction	650	Sanidhya Gupta, Sanidhay Bhambay, Narges Alavisamani, Neil Walton and Thirupathiah Vasantam	Boundary-Aware Stabilizer Scheduling for Distributed Quantum Error Correction
25	2-TUE	15:00 -16:30	604	QNET	QNET::650::708::710	Distributed Quantum Computing & Error Correction	708	Nitish Kumar Chandra, Reza Nejabati and Eneet Kaur	Towards the Characterization of Logical Errors in Distributed Lattice Surgery
26	2-TUE	15:00 -16:30	604	QNET	QNET::650::708::710	Distributed Quantum Computing & Error Correction	710	Stav Haldar, Filip Rozpedek and Sumeet Khatri	Hardware-informed entanglement distribution for distributed quantum computing using reinforcement learning

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27	3-WED	10:00 -11:30	604	QNET	QNET::194::489::1261	Quantum Network Control & Switching	194	Alexander Pirker, Robert Hayek, Alexander Kolar, Igor Kadota, Joaquin Chung and Rajkumar Kettimuthu	Centralizing Task-based Approach to Quantum Network Control
28	3-WED	10:00 -11:30	604	QNET	QNET::194::489::1261	Quantum Network Control & Switching	489	Aman Yacob Tekleab, Yifeng Shen, Yutaro Yoshii, Amin Taherkani, Rodney Van Meter and Shota Nagayama	A Distributed Switching Protocol for Quantum Networks
29	3-WED	10:00 -11:30	604	QNET	QNET::194::489::1261	Quantum Network Control & Switching	1261	Jake Smith, Thomas Beauchamp, Scarlett Gauthier, Oumayma Bouchmal and Stephanie Wehner	Tools for Reducing Service Time in Near-Term Quantum Network
30	3-WED	13:00 -14:30	604	QNET	QNET::254::557::883	Quantum Repeater Performance & Resource Allocation	254	Karim S. Elsayed and Amr Rizk	Quantum Memory Allocation for High Fidelity Asymmetric Quantum Repeaters
31	3-WED	13:00 -14:30	604	QNET	QNET::254::557::883	Quantum Repeater Performance & Resource Allocation	557	Van Sy Mai, Abderrahim Amlou, Richard La, Cory Nunn and Abdella Battou	On the Swapping Capacity of a Quantum Repeater
32	3-WED	13:00 -14:30	604	QNET	QNET::254::557::883	Quantum Repeater Performance & Resource Allocation	883	Sagar Patange, Caitao Zhan, Bikun Li, Joaquin Chung, Allen Zang, Liang Jiang and Rajkumar Kettimuthu	Realistic Simulation of Quantum Repeater with Encoding and Classical Error Correction
33	3-WED	15:00 -16:30	604	QNET	QNET::7::1071::431	Quantum Authentication & Post-Quantum Protocols	7	Ganesh Ramachandran Nair and Tony Thomas	A Measurement-Time Quantum Integrity Scheme for Terminal Protocols
34	3-WED	15:00 -16:30	604	QNET	QNET::7::1071::431	Quantum Authentication & Post-Quantum Protocols	1071	Kevin Bogner, Aysajan Abidin, Dave Singelee and Bart Preneel	Tight Quantum Distance-Bounding with Conditional Terrorist-Fraud Resistance
35	3-WED	15:00 -16:30	604	QNET	QNET::7::1071::431	Quantum Authentication & Post-Quantum Protocols	431	Nadeem Ahmed, Aryya Gangopadhyay and Lei Zhang	NoisePQC++: A Unified NIST-Compliant PQC and Hybrid-PQC Implementation of the Noise Protocol
36	4-THU	10:00 -11:30	604	QNET	QNET::749::1103::1239	Quantum Data Center Networks & Secure Infrastructure	749	Jiyao Liu, Lei Fan, Yuanxiong Guo, Zhu Han and Yu Wang	Topology Optimization in All-Optical Switched Quantum Data Center Networks
37	4-THU	10:00 -11:30	604	QNET	QNET::749::1103::1239	Quantum Data Center Networks & Secure Infrastructure	1103	Joseph Meyer, Juan Jose Vegas Olmos and Idelfonso Tafur Monroy	QKD Post-Processing Offload on BlueField-3 DPUs: Towards Quantum-Secured RDMA in Datacenter Fabrics
38	4-THU	10:00 -11:30	604	QNET	QNET::749::1103::1239	Quantum Data Center Networks & Secure Infrastructure	1239	Mariam Gado and Muhammad Ismail	Multi-Year Power Upgrades to Support SQKD with Heterogeneous Quantum Servers in a Pre-Existing Cyber Layer Under Rising Key Rate Demands
39	4-THU	10:00 -11:30	605	QNET	QNET::735::1108::1173	Distributed Quantum Computing & QDC Architectures	735	Samanvay Sharma, Siyuan Niu, Rodney Van Meter, Amin Taherkhani and Michal Hajdušek	Performance Analysis of QAOA Across Distributed Quantum Network Topologies Using SwitchQNet
40	4-THU	10:00 -11:30	605	QNET	QNET::735::1108::1173	Distributed Quantum Computing & QDC Architectures	1108	Chien-Tung Kuo, Felix Burt, Samuel Yen-Chi Chen, Kin K. Leung and Kuan-Cheng Chen	Architecture-Aware Reinforcement Learning for Communication-Efficient Distributed Quantum Circuit Compilation
41	4-THU	10:00 -11:30	605	QNET	QNET::735::1108::1173	Distributed Quantum Computing & QDC Architectures	1173	Shahrooz Pouryousef, Eneet Kaur, Hassan Shapourian, Don Towsley, Ramana Kompella and Reza Nejabati	Benchmarking Quantum Data Center Architectures: A Performance and Scalability Perspective
42	4-THU	13:00 -14:30	604	QNET	QNET::302::220::314	Classical-Control Security & Physical-Layer Side Channels	302	Ali Hamza Malik, Raja Hasnain Anwar and Muhammad Taqi Raza	Beyond the Quantum Promise: A Security Analysis of Classical Control in Quantum Key Distribution
43	4-THU	13:00 -14:30	604	QNET	QNET::302::220::314	Classical-Control Security & Physical-Layer Side Channels	220	Marwan Elawady, Lance Young, Contessa Wilburn, Blaine Keyton, Carrie Houston, Mohamed Shaban and Muhammad Ismail	Inferring Protocol Execution Stages in Quantum Networks via Physical Layer Side Channel Analysis
44	4-THU	13:00 -14:30	604	QNET	QNET::302::220::314	Classical-Control Security & Physical-Layer Side Channels	314	Lance Young, Contessa Wilburn, Carrie Houston, Blaine Keyton, Marwan Elawady, Mohamed Shaban and Muhammad Ismail	Protocol Fingerprinting in Quantum Networks via Physical Layer Side Channel Analysis
45	4-THU	15:00 -16:30	604	QNET	QNET::895::1120::1170	PQC Migration & Quantum-Safe Applications	895	Matthew Chou and Phuong Cao	Network Impact of Post-Quantum Certificate Chain sizes on Time to First Byte in TLS Deployments
46	4-THU	15:00 -16:30	604	QNET	QNET::895::1120::1170	PQC Migration & Quantum-Safe Applications	1120	Dhruv Bhatnagar and Amandeep Bhatia	Quantum Image Scrambling Algorithm Based on Discrete Baker Map and Cosine Transformation
47	4-THU	15:00 -16:30	604	QNET	QNET::895::1120::1170	PQC Migration & Quantum-Safe Applications	1170	Ahmeed Yinusa, Khem Poudel, Misa Faezipour and Clifford Jones	Post-Quantum Medical Imaging Security: A Survey of Architectures and Migration Trade-Offs

QECS - Quantum End-to-End Case Studies

QCE26 Technical Papers Track

v123	Day	Time	Room	Track	QCE-PRG-ID	Session Name	ECID	Paper Authors	Paper Title
1	1-MON	10:00 -11:30	802A/B	QECS	QECS::629::968::939	Biology & Chemistry	629	Guilherme Cruz, Nouhaila Innan, Alberto Marchisio, Gabriel Falcao and Muhammad Shafique	Q-Blood: Enhancing Blood Cells Classification using Hybrid Quantum Neural Networks
2	1-MON	10:00 -11:30	802A/B	QECS	QECS::629::968::939	Biology & Chemistry	968	Alejandro Gomez Cadavid, Pavle Nikačević, Pranav Chandarana, Sebastian Romero, Enrique Solano, Narendra Hegade, Miguel Angel Lopez-Ruiz, Claudio Grotto, Hanna Linn, Hakan Doga, Evgeny Epifanovsky, Panagiotis Kl. Barkoutsos, Ananth Kaushik and Martin Roetteler	Protein folding on a 64 qubit trapped-ion hardware via counterdiabatic quantum optimization
3	1-MON	10:00 -11:30	802A/B	QECS	QECS::629::968::939	Biology & Chemistry	939	Marek Kowalik, Ellen Michael, Peter Pogány and Phalgun Lolur	Practical Scalability of Tensor Network Quantum Emulators for Molecular Hamiltonian Simulation
4	1-MON	13:00 -14:30	802A/B	QECS	QECS::863::1030::1162	Logistics & Supply Chain I	863	Marek Kowalik, Sam Genway, Vedangi Pathak, Mykola Maksymenko, Simon Martiel, Hamed Mohammadbagherpoor, Richard Padbury, Vladyslav Los, Oleksa Hryniv, Peter Pogány and Phalgun Lolur	Executing Utility-Scale Hamiltonian Dynamics for Covalent Inhibitor Reactivity on Quantum Hardware
5	1-MON	13:00 -14:30	802A/B	QECS	QECS::863::1030::1162	Logistics & Supply Chain I	1030	Jiri Guth Jarkovsky, Patricia Bickert, Elisabeth Wybo and Martin Leib	Rolling Stock Planning Using the Quantum Approximate Optimization Algorithm
6	1-MON	13:00 -14:30	802A/B	QECS	QECS::863::1030::1162	Logistics & Supply Chain I	1162	Camille de Valk, Koen Reerink, Siert Sebus and Sébastian de Bone	Hybrid quantum-classical end-to-end pipeline for solving MILPs: a vehicle routing case study
7	1-MON	15:00 -16:30	802A/B	QECS	QECS::400::898::313	Climate & Environment	400	Lea Casse, Sabarikirishwaran Ponnambalam, Gregory Pearson, Nick Lim, Bernhard Pfahringer and Albert Bifet	Hybrid Quantum Risk Minimisation: A QRU-QAOA Pipeline for Spatial Flood Tail-Risk Allocation
8	1-MON	15:00 -16:30	802A/B	QECS	QECS::400::898::313	Climate & Environment	898	Tom Montgomery, Adrian Varela-Alvarez Varela-Alvarez, Sam Genway and Phalgun Lolur	Accelerating MOF Discovery for Climate Change Mitigation: A Hierarchical Quantum Embedding Approach
9	1-MON	15:00 -16:30	802A/B	QECS	QECS::400::898::313	Climate & Environment	313	Samar Aseeri, Jeff Burgett, Julián Stiefkens, Tom Taylor and Adam Milewski	VQ-MAR: A Variational Quantum–Classical Hybrid Framework for Managed Aquifer Recharge Site Selection
10	2-TUE	10:00 -11:30	713A	QECS	QECS::1130::685::368	Optimization & Simulation	1130	Prashanti Priya Angara, Luis F. Rivera, Ulrike Stege, Hausi Müller, Ibrahim Shehzad and Sean Wagner	Experimental Workflows for Combinatorial Optimization: Towards Quantum Advantage
11	2-TUE	10:00 -11:30	713A	QECS	QECS::1130::685::368	Optimization & Simulation	685	Jorge Saavedra, Jhon Alejandro Montanez, Alberto Maldonado and Daniel Sierra-Sosa	Hybrid Classical-Quantum Approach for Multi-Constrained Location Optimization Problem
12	2-TUE	10:00 -11:30	713A	QECS	QECS::1130::685::368	Optimization & Simulation	368	Rushit Kansara and Maria Isabel Roldan Serrano	Quantum Assisted Genetic Algorithm for Constrained Multi-Objective Optimization Problems
13	2-TUE	13:00 -14:30	713A	QECS	QECS::412::308::814	Algorithms & Performance	412	Daiwei Zhu, Miguel Angel Lopez-Ruiz, François-Henry Rouet, Claudio Grotto, Willie Aboumrاد, Robert Lucas, Ananth Kaushik and Martin Roetteler	End-to-end performance of quantum-accelerated large-scale linear algebra workflows
14	2-TUE	13:00 -14:30	713A	QECS	QECS::412::308::814	Algorithms & Performance	308	Kevin Dougherty and Ji Liu	A Unified Framework for Operator Backpropagation and Observable Measurement in Quantum Computing
15	2-TUE	13:00 -14:30	713A	QECS	QECS::412::308::814	Algorithms & Performance	814	Zefan Du, Zihao Jiang, Shaolun Ruan, Juntao Chen, Yong Wang, Long Cheng, Rajkumar Buyya and Ying Mao	Resource-Efficient and Self-Adaptive Quantum Search in a Quantum-Classical Hybrid System
16	2-TUE	15:00 -16:30	713A	QECS	QECS::418::937::865	Logistics & Supply Chain II	418	Miguel Angel Lopez-Ruiz, Daiwei Zhu, Jonas Hatzenbuehler, Shudian Zhao, Claudio Grotto, Willie Aboumrاد, Jonas Alm, Julia Kompalla, Mena Issler, Ananth Kaushik and Martin Roetteler	Hybrid Quantum-Classical Optimization Workflows for the Shipment Selection Problem
17	2-TUE	15:00 -16:30	713A	QECS	QECS::418::937::865	Logistics & Supply Chain II	937	Chia-Ho Ou, Chia-Chen Yang, Masayuki Ohzeki and Chih-Yu Chen	Hybrid Reverse Annealing for Carbon-Efficient Truck-Drone Delivery via Scalable QUBO Decomposition
18	2-TUE	15:00 -16:30	713A	QAPP	QECS::418::937::865	Quantum Error Mitigation, Fidelity & Verification	865	Farrokh Labib, David Nicholaeff, Vincent Russo and William Zeng	Compressed Sensing for Efficient Fidelity Estimation of GHZ States
19	3-WED	10:00 -11:30	703	QECS	QECS::435::301::660	Optimization & Simulation	435	Nicholas DiBrita, Jason Han, Krishna Bhatia, Younghyun Cho, Hengrui Luo and Tirthak Patel	Domain-Aware Probability Sampling for Hybrid Quantum Systems using Bayesian Optimization
20	3-WED	10:00 -11:30	703	QECS	QECS::435::301::660	Optimization & Simulation	301	Zack Hassman, Oliver Reardon-Smith, Gokul Subramanian Ravi, Frederic Chong and Kevin Sung	Extraferm: An Extended Matchgate Simulator
21	3-WED	10:00 -11:30	703	QECS	QECS::435::301::660	Optimization & Simulation	660	Arul Mazumder	An End-to-End Hybrid Quantum–Classical Sampling Workflow for Discrete Markov Random Fields: A Reproducible Case Study

QPHO - Quantum Photonics
CSE26 Techical Papers Track

v123	Day	Time	Room	Track	QCE-PRG-ID	Session Name	ECID	Paper Authors	Paper Title
1	0-SUN	10:00 -11:30	713A	QPHO	QPHO::824::885::981	Photonic Quantum Information: Measurement, Characterization, and Applications	824	Mitchell Thornton, Hiva Shahoei, Cooper Shapard, R. Mason Tuller and Duncan MacFarlane	Programmable Basis-Selective Measurement for Photonic Qudits Using a Single-Qudit Gate
2	0-SUN	10:00 -11:30	713A	QPHO	QPHO::824::885::981	Photonic Quantum Information: Measurement, Characterization, and Applications	885	Nirvik Sahoo, Paul R Griffin and Chyng Wen Tee	Uncovering Asset Manager Utility and Market Risk via a Hybrid Photonic Quantum Reinforcement Learning Framework
3	0-SUN	10:00 -11:30	713A	QPHO	QPHO::824::885::981	Photonic Quantum Information: Measurement, Characterization, and Applications	981	Anuvab Sen and Saibal Mukhopadhyay	SSP-QST: Spectral Subspace Purification for Photonic Quantum State Tomography
4	0-SUN	13:00 -14:30	713A	QPHO	QPHO::1240::1244	Advances in Quantum Nanophotonics and High-Resolution Light Detection	1240	Nur Duwi Fat Fitri, Suseong Lim, Jin-Woo Kim, Heonoh Kim and June-Koo Kevin Rhee	Enhancing Resolution of Spatially Resolved Quantum Imaging of Two-Photon Interference By Multiphoton Thresholding With EMCCD
5	0-SUN	13:00 -14:30	713A	QPHO	QPHO::1240::1244	Advances in Quantum Nanophotonics and High-Resolution Light Detection	1244	Julien Werzowa, Li Liu and Viktoriia Babicheva	Resonant Control of Quantum Emitters with Super-Mossian van der Waals Nanoantennas
6	0-SUN	15:00 -16:30	713A	QPHO	QPHO::446::1134	Engineering Scalable Quantum Photonic Integrated Circuits	446	Gabriele De Angelis, Nicolò Leone, Alessandro Luongo, Alberto Montanaro, Matteo Sanna, Roberto Siagri, Vito Sorianello, Luigi Tallone, Fabrizio Tamburini and Marco Venere	Design and Benchmarking of a Quantum Photonic Chip
7	0-SUN	15:00 -16:30	705	QPHO	QPHO::446::1134	Engineering Scalable Quantum Photonic Integrated Circuits	1134	Ryota Kitagawa, Yeting Yang, Toshiaki Iwai, Tetsuya Miyatake, Masaharu Hida, Koki Kaminaka, Naoki Fushimi, Takuto Yamaguchi, Tetsuro Ishiguro, Itsuki Takagi, Masayuki Hosoda, Hidetsugu Matsukiyo, Satomi Ishida, Teruya Ishihara, Satoshi Iwamoto, Manabu Ohtomo, Toshiyuki Miyazawa, Kenichi Kawaguchi, Ryoichi Ishihara and Shintaro Sato	Fabrication and Screening of Quantum Photonic Chip with Single Diamond Tin-Vacancy Centers towards Scalable Quantum Computing
8	2-TUE	10:00 -11:30	605	QPHO	QPHO::1140::241::256	Photonic Quantum Systems: Synthesis, Calibration, and Compilation	1140	Tingxiang Ji, Hansika Weerasena, Demitry Farfurnik and Jianqing Liu	Towards Efficient Synthesis of Quantum Graph States by Fusing Graph Motifs
9	2-TUE	10:00 -11:30	605	QPHO	QPHO::1140::241::256	Photonic Quantum Systems: Synthesis, Calibration, and Compilation	241	Romain Piron and Akihito Soeda	Parameter Calibration for Reduced-Bandwidth Two-Photon Waveguide-QED Simulations
10	2-TUE	10:00 -11:30	605	QPHO	QPHO::1140::241::256	Photonic Quantum Systems: Synthesis, Calibration, and Compilation	256	Tobias Forster, Yannick Stade, Andreas Fyrrillas, Jean Senellart, Lukas Burgholzer and Robert Wille	Handling Imperfections in Subcircuit Compilation for Linear Optical Quantum Computing

QML - Quantum Machine Learning

QCE26 Techical Papers Track

v123	Day	Time	Room	Track	QCE-PRG-ID	Session Name	ECID	Paper Authors	Paper Title
v123	0-SUN	10:00 -11:30	803A	QML	QML::1242::825::877	Quantum Kernels for Anomaly Detection	1242	Kauã Barbosa, Everson Flores, Bruna Guterres, Silvia Botelho, and Marcelo Pias	Unsupervised Quantum Kernels for One-Class Fault Detection in Industrial Time Series
2	0-SUN	10:00 -11:30	803A	QML	QML::1242::825::877	Quantum Kernels for Anomaly Detection	825	Takao Tomono and Kazuya Tsujimura	Identification and Classification of Ten Individual Anomaly Types using Quantum Kernel One-Class SVM with Non-Contact Acoustic Sensing
3	0-SUN	10:00 -11:30	803A	QML	QML::1242::825::877	Quantum Kernels for Anomaly Detection	877	Hyong Keun Park, Joonkyo Shim and Hyunsoo Yoon	Protocol Sensitivity in Quantum Feature Augmentation for Industrial Anomaly Detection
4	0-SUN	13:00 -14:30	803A	QML	QML::1138::186::520	Adversarial Robustness of Quantum Classifiers	1138	Vishwa Hatan Achchige, Kasun Gunawardana, Asitha Kottahachchi Kankanange Don, Charitha Elvitigala and Ibrahim Khalil	Inference Shock and Noise Placement in Hybrid Quantum Neural Networks
5	0-SUN	13:00 -14:30	803A	QML	QML::1138::186::520	Adversarial Robustness of Quantum Classifiers	186	Emma Andrews, Sahan Sanjaya and Prabhat Mishra	Defending Quantum Classifiers against Adversarial Perturbations through Quantum Autoencoders
6	0-SUN	13:00 -14:30	803A	QML	QML::1138::186::520	Adversarial Robustness of Quantum Classifiers	520	Emma Andrews, Nahyeon Kim and Prabhat Mishra	Quantum Interval Bound Propagation for Certified Training of Quantum Neural Networks
7	0-SUN	15:00 -16:30	803A	QML	QML::338::444::1174	Combinatorial Optimization & QAOA	338	Kien Nguyen and Ilya Safro	Graph-Conditioned Meta-Optimizer for QAOA Parameter Generation on Multiple Problem Classes
8	0-SUN	15:00 -16:30	803A	QML	QML::338::444::1174	Combinatorial Optimization & QAOA	444	Daeyeun Kim, Seungcheol Oh, Yun Pyo Hong and Joongheon Kim	Coordinate Descent for Multi-Angle QAOA
9	0-SUN	15:00 -16:30	803A	QML	QML::338::444::1174	Combinatorial Optimization & QAOA	1174	Zirui Zhang, Pravin Mahendran, Kevin Zhang and David E. Bernal Neira	Towards Maximum Independent Set Inspired Dataset Reductions with Neutral Atom Quantum Computers
10	1-MON	10:00 -11:30	703	QML	QML::237::726::988	Applications & Security	237	Abdallah Aaraba, Alexis Vieloszynski, Remon Polus, Ola Ahmad and Soumaya Cherkaoui	RF Spectrogram Anomaly Detection with Quantum Kitchen Sinks: Architecture, Representation, and Hardware Validation
11	1-MON	10:00 -11:30	703	QML	QML::237::726::988	Applications & Security	726	Junrui Zhang, Zemin Chen, Lusi Li, Mohammad Ghasemigol, Daniel Takabi and Rui Ning	Input-Aware Dynamic Backdoor Attack Against Quantum Neural Networks
12	1-MON	10:00 -11:30	703	QML	QML::237::726::988	Applications & Security	988	Taha Gharaibeh and Jyotsna Sharma	Small Gates, Big Gains:Quantum Feature Pyramid Gating for Seismic Image Segmentation
13	2-TUE	10:00 -11:30	803A	QML	QML::286::978::229	Quantum Architecture & Encoding Search	286	Swagat Kumar, Jan-Nico Zaech, Colin Wilmott and Luc Van Gool	Simulating Mixed State Dynamics to Model Differentiable Quantum Architecture Search
14	2-TUE	10:00 -11:30	803A	QML	QML::286::978::229	Quantum Architecture & Encoding Search	978	Nadine Charaf, Mahdi Chehimi, Samuel Yen-Chi Chen and Abdallah Shami	Differentiable Quantum Architecture Search for HEP Data Classification
15	2-TUE	10:00 -11:30	803A	QML	QML::286::978::229	Quantum Architecture & Encoding Search	229	Lena Tokuhiro, Amine Bentellis and Jeanette Miriam Lorenz	Discovering Data Encoding Strategies for Quantum-Classical Neural Networks Using Monte Carlo Tree Search
16	2-TUE	10:00 -11:30	705	QML	QML::1025::299::723	Quantum Feature Representations	1025	Daniel Fadi Feghali, Hasan Almatrouk, Kin Tung Michael Ho, Kuan-Cheng Chen, Lily Lee, Jen-Yu Chang and Po-Heng Lee	Quantum Kernel Feature Selection via Quadratic Unconstrained Binary Optimization
17	2-TUE	10:00 -11:30	705	QML	QML::1025::299::723	Quantum Feature Representations	299	Andras Ferenczi, Dagen Wang, Mariya Bessonova, Sutapa Samanta, Todd Hodges, John Hancock, Guillermo Vilarino, Amol Deshmukh, Mariana LaDue, Girish Pillai and Hilary Packer	Credit Default Prediction with Projected Quantum Feature Models and Ensembles
18	2-TUE	10:00 -11:30	705	QML	QML::1025::299::723	Quantum Feature Representations	723	Sang Hyub Kim, Oliver Knitter, Jonathan Mei, Claudio Grotto, Masako Yamada, Martin Roetteler and Chi Chen	Quantum Parity Representations: Learnable Basis Discovery, Encoders, and Shadow Deployment
19	2-TUE	13:00 -14:30	803A	QML	QML::324::462::447	Quantum ML for Chemistry & Molecules	324	James T. Pegg, Hubert Okadome Valencia and Ronin Wu	Quantum and Classical Implementations of Topology-Aligned Architectures for Molecular Property Prediction
20	2-TUE	13:00 -14:30	803A	QML	QML::324::462::447	Quantum ML for Chemistry & Molecules	462	Abdul Hannan, Maral Mahmoudi Kamelabad, Anna Kazakova Lindegren and Robert Loredó	Quantum Feature Mapping for Molecular Classification: Evaluating Exponential Expressivity in NISQ Hardware
21	2-TUE	13:00 -14:30	803A	QML	QML::324::462::447	Quantum ML for Chemistry & Molecules	447	Shunji Matsuura and Sonika Johri	Discretization-Aware Fine-Tuning for Quantum Machine Learning with Chemical Foundation Models
22	2-TUE	15:00 -16:30	803A	QML	QML::1209::640::142	Quantum ML for Healthcare	1209	Mehtap Agirsoy, Batuhan Yalcin and Matthew A. Oehlschlaeger	variational Quantum Classifiers for In-vitro Fertilisation Pregnancy Outcome Prediction: Benchmarking VQC and QNN against Classical Baselines on Ideal Noisy and Real Quantum Hardware
23	2-TUE	15:00 -16:30	803A	QML	QML::1209::640::142	Quantum ML for Healthcare	640	Sydney Leither, Thomas Lubinski, Michael Kubal and Sonika Johri	Benchmarking Quantum and Classical Machine Learning Models on Oncological Data
24	2-TUE	15:00 -16:30	803A	QML	QML::1209::640::142	Quantum ML for Healthcare	142	Sachin Namdeo, Henry Sukumar, Asvija B. and Bijoy M. B.	Parameter-Efficient Quantum-Classical Hybrid Network for Edge-Based Medical Imaging Systems
25	3-WED	10:00 -11:30	803A	QML	QML::470::509::119	Quantum Reinforcement Learning	470	Mohamad Khair Altrabulsi, Nouhaila Innan, Alberto Marchisio, Muhammad Kashif and Muhammad Shafique	Q-SpiRL: Quantum Spiking Network Reinforcement Learning for Adaptive Robot Navigation
26	3-WED	10:00 -11:30	803A	QML	QML::470::509::119	Quantum Reinforcement Learning	509	Salvatore Zammuto and Martin Schulz	Quantum-Classical Reinforcement Learning for Molecular Excitation-Gap Engineering

QML - Quantum Machine Learning

QCE26 Techical Papers Track

27	3-WED	10:00 -11:30	803A	QML	QML::470::509::119	Quantum Reinforcement Learning	119	Hyojun Ahn, Hoyeong Lee, Soohyun Park, Soyi Jung, Minseok Choi, Kaigui Bian and Joongheon Kim	AQuA-Cache: Agentic Quantum-AI for LLM-Guided Model Selection in Large-Scale Short-Form Video Services
28	3-WED	13:00 -14:30	803A	QML	QML::1228::876::387	Quantum Transformers & Multimodal Learning	1228	Srikar Alla and Ali Shiri Sichani	Quantum Entangled Multimodal Fusion Networks (QEMFN): Resource-Aware Hybrid Vision-Language Fusion via Trainable Entanglement
29	3-WED	13:00 -14:30	803A	QML	QML::1228::876::387	Quantum Transformers & Multimodal Learning	876	Mina Abbaszadeh, Matilda Karabina Moore, Mehrnoosh Sadrzadeh and Martha Lewis	Quantum Models with Multi-Stage Training for Compositional Concept Generalization
30	3-WED	13:00 -14:30	803A	QML	QML::1228::876::387	Quantum Transformers & Multimodal Learning	387	Andrea Gasparini and Diego Iacopetta	Watching Quantum Models Think: Hilbert-Space Interpretability in Quantum Transformer Blocks
31	3-WED	15:00 -16:30	803A	QML	QML::1113::1178::253	Quantum State Learning & Error Correction	1113	Kashyap Patel, Bhavin Makwana and Manjunath Joshi	Bloch–Wasserstein Generative Adversarial Network for Quantum State Learning
32	3-WED	15:00 -16:30	803A	QML	QML::1113::1178::253	Quantum State Learning & Error Correction	1178	Anders Buvarp and Stefan Werner	Toward robust real-time bosonic decoding with geometry-aware neural networks
33	3-WED	15:00 -16:30	803A	QML	QML::1113::1178::253	Quantum State Learning & Error Correction	253	Nico Meyer, Christopher Mutschler, Dominik Seuß, Andreas Maier and Daniel D. Scherer	Learning to Concatenate Quantum Codes
34	4-THU	10:00 -11:30	803A	QML	QML::334::424::1139	Sequential & Temporal Quantum Learning	334	Kuo-Chung Peng, Samuel Yen-Chi Chen, Jiun-Cheng Jiang, Chen-Yu Liu, En-Jui Kuo, Yun-Yuan Wang, Prayag Tiwari, Andrea Ceschini, Chi-Sheng Chen, Yu-Chao Hsu, Chun-Hua Lin, Tai-Yue Li, Antonello Rosato, Massimo Panella, Simon See, Saif Al-Kuwari, Kuan-Cheng Chen, Nan-Yow Chen and Hsi-Sheng Goan	Gated QKAN-FWP: Scalable Quantum-inspired Sequence Learning
35	4-THU	10:00 -11:30	803A	QML	QML::334::424::1139	Sequential & Temporal Quantum Learning	424	Samuel Yen-Chi Chen, Yifeng Peng, Kuo-Chung Peng, Jiun-Cheng Jiang, Chun-Hua Lin, Junghoon Justin Park, Huan-Hsin Tseng, Hsin-Yi Lin, Kuan-Cheng Chen, Chen-Yu Liu and Shinjae Yoo	Self-Modulating Quantum Fast-Weight Programmers for Efficient Adaptive Sequential Learning
36	4-THU	10:00 -11:30	803A	QML	QML::334::424::1139	Sequential & Temporal Quantum Learning	1139	Krishna Bhatia, Bhavin Makwana, Manjunath Joshi and Shalini Devendrababu	Quantum Reservoir Computing with a Temporal Convolutional Readout for Optical Equalization and Forecasting
37	4-THU	13:00 -14:30	803A	QML	QML::1041::793::248	Fourier & Harmonic Quantum Models	1041	Melvin Strobl, Maja Franz, Lukas Scheller, Eileen Kuehn, Wolfgang Mauerer and Achim Streit	Beyond Gates: Pulse Level Quantum Fourier Models
38	4-THU	13:00 -14:30	803A	QML	QML::1041::793::248	Fourier & Harmonic Quantum Models	793	Michael Poppel, David Bucher, Maximilian Zorn, Nico Kraus, Claudia Linnhoff-Popien, Philipp Altmann and Jonas Stein	Mitigating Exponential Mixed Frequency Growth in QML
39	4-THU	13:00 -14:30	803A	QML	QML::1041::793::248	Fourier & Harmonic Quantum Models	248	Kristina Kirova, Monika Doerfler, Franz Luef and Richard Kueng	Harmoniq: Efficient Data Augmentation on a Quantum Computer Inspired by Harmonic Analysis
40	4-THU	15:00 -16:30	803A	QML	QML::510::908	Federated Quantum Learning	510	Walid El Maouaki and Muhammad Shafique	Q-RAIL: A Reliability-Aware Framework for Quantum Federated Learning on Heterogeneous Noisy Hardware
41	4-THU	15:00 -16:30	803A	QML	QML::510::908	Federated Quantum Learning	908	Fabrice Kameni, Md Martuza Ahamad, Md Munirul Haque and Sheikh Iqbal Ahamed	Noise-Aware and Shot-Adaptive Training for Federated Variational Quantum Learning
42	5-FRI	10:00 -11:30	705	QML	QML::866::671::900	Generalization & Efficient Quantum Models	866	Jun Towada	Circuit Depth, Boundary Complexity, and Generalization in Few-Data Local VQCs
43	5-FRI	10:00 -11:30	705	QML	QML::866::671::900	Generalization & Efficient Quantum Models	671	Van Tien Nguyen and Panagiotis Markopoulos	VQBR: Variational Quantum Bayesian Regression via Measurement-Based MAP Direction Recovery
44	5-FRI	10:00 -11:30	705	QML	QML::866::671::900	Generalization & Efficient Quantum Models	900	Li Zhong, Sichen Zhang, Fabien Friedel, Julian Seyfarth, Lorenz John and Jan Wehinger	QGSP: Quantum Graph Spectral Positioning for Graph Representation Learning
45	5-FRI	13:00 -14:30	705	QML	QML::1154::532::1268	Trainability & Barren Plateaus	1154	Andrew Barlow, Markus Lange and Hans-Martin Rieser	Employing Controllable Quantum Circuits for use in Quantum Machine Learning
46	5-FRI	13:00 -14:30	705	QML	QML::1154::532::1268	Trainability & Barren Plateaus	532	Yichen Xie, Cassandre Notton and Jean Senellart	Pre-Asymptotic Trainability in Photonic Variational Circuits under Postselection
47	5-FRI	13:00 -14:30	705	QML	QML::1154::532::1268	Trainability & Barren Plateaus	1268	Junyong Lee, Jeihee Cho, Daniel Justice, Euimin Lee, Hyeonseong Jung and Shiho Kim	Variational Quantum Hypernetworks for Generalized Quantum Circuit Initialization
48	5-FRI	15:00 -16:30	705	QML	QML::639::838::452	Quantum ML for the Physical Sciences	639	Nahid Binandeh Dehaghani, Ban Q. Tran, Susan Mengel, Rafal Wisniewski and A. Pedro Aguiar	A QPINN Framework with Quantum Trainable Embeddings for the Lid-Driven Cavity Problem
49	5-FRI	15:00 -16:30	705	QML	QML::639::838::452	Quantum ML for the Physical Sciences	838	Qimao Yang and Jing Guo	Quantum Hypernetwork Method for Semiconductor Process Simulation
50	5-FRI	15:00 -16:30	705	QML	QML::639::838::452	Quantum ML for the Physical Sciences	452	Vasile Vancea	Biologically-Constrained Quantum Encoding and Quantum Convolutional Neural Networks for Modeling DNA Strand Displacement Kinetics

QTEM - Quantum Technologies and Systems Engineering

QCE26 Technical Papers Track

v123	Day	Time	Room	Track	QCE-ID	Session Name	ECID	Paper Authors	Paper Title
1	0-SUN	10:00 -11:30	802A/B	QTEM	QTEM::363::839::318	Quantum Control, Optimization & Reinforcement Learning I	363	Alexander Leong	Algebraic Reduction and Numerical Optimization Methods
2	0-SUN	10:00 -11:30	802A/B	QTEM	QTEM::363::839::318	Quantum Control, Optimization & Reinforcement Learning I	839	Qimao Yang, Hanbo Yang and Jing Guo	Co-Design of Quantum Hardware and Control Using Reinforcement Learning
3	0-SUN	10:00 -11:30	802A/B	QTEM	QTEM::363::839::318	Quantum Control, Optimization & Reinforcement Learning I	318	Andrew Kamen, Samuel Fine, Bikrant Bhattacharyya, Frederic Chong and Andy Goldschmidt	Comparing and Correcting Robustness Metrics for Quantum Optimal Control
4	0-SUN	13:00 -14:30	802A/B	QTEM	QTEM::488::404::1264	Quantum Error Correction & Fault Tolerance	488	Leo Itoh, Takuya Kasamura, Hidetsugu Irie and Junichiro Kadomoto	Design of a Quantum Error Correction Decoder Exploiting Temporal Parallelism
5	0-SUN	13:00 -14:30	802A/B	QTEM	QTEM::488::404::1264	Quantum Error Correction & Fault Tolerance	404	Konstantinos-Nikolaos Papadopoulos, Kaitlin N. Smith and Jakub Szefer	On the Feasibility of Time-Multiplexed Control for Surface Code Fault Tolerant Computation
6	0-SUN	13:00 -14:30	802A/B	QTEM	QTEM::488::404::1264	Quantum Error Correction & Fault Tolerance	1264	Jan-Erik R. Wichmann, Werner Florian Samayoa, Prasoon Ambalathankandy and Kentaro Sano	Scalable FPGA-based Decoder for Quantum Error Correction with Syndrome Subgraph Algorithm
7	0-SUN	15:00 -16:30	802A/B	QTEM	QTEM::528::383::416	Quantum Algorithms, Architecture & Systems	528	Mason Edwards and Prabhat Mishra	Branch-Resolved Characterization of Feed-Forward Error in Dynamic Teleportation via Classical Choi Shadows
8	0-SUN	15:00 -16:30	802A/B	QTEM	QTEM::528::383::416	Quantum Algorithms, Architecture & Systems	383	Jason Ludmir and Tirthak Patel	PaQit: Energy-Runtime-Fidelity Co-Optimization for Neutral Atom Quantum Computers
9	0-SUN	15:00 -16:30	802A/B	QTEM	QTEM::528::383::416	Quantum Algorithms, Architecture & Systems	416	Aishwarya Majumdar and Yuan Liu	Robust Quantum Algorithmic Binary Decision-Making on Gaussian Signals
10	1-MON	10:00 -11:30	605	QTEM	QTEM::310::958::1069	Emerging Devices & Characterization	310	Nathan Jones, Binayyak Roy, Valentine Mohaugen, Ian Lewis, Toby Cox, Sumanta Tewari and Rong Ge	MEDA: Measurement-Efficient Disorder-Aware Detection of Majorana Zero Modes in Realistic Devices
11	1-MON	10:00 -11:30	605	QTEM	QTEM::310::958::1069	Emerging Devices & Characterization	958	Sakineh Ghaderi, Abdelkrim Menina, Leon Röscher, Talía L. M. Lezama, Riccardo Bassoli and Frank H.P.Fitzek	Scheduling Under Stochastic Control Drift in a Room-Temperature NV-Center Quantum Processor
12	1-MON	10:00 -11:30	605	QTEM	QTEM::310::958::1069	Emerging Devices & Characterization	1069	Tobias Schmalzried, Jens Wizl, Valentyna Afanasenko, Diego Kuderna Melgar and Ingmar Kalfass	Sparse-Point-Enabled full-Spectrum Reconstruction for Rapid NV-ODMR Data Acquisition
13	4-THU	10:00 -11:30	703	QTEM	QTEM::1226::666::523	Superconducting Qubit Hardware & Packaging	1226	Shreyan Juvvadi, Sarah Dweik, Likai Yang and Hiu Yung Wong	3D Simulation of Flip Chip Qubit System with Intermediate Coupling Layer
14	4-THU	10:00 -11:30	703	QTEM	QTEM::1226::666::523	Superconducting Qubit Hardware & Packaging	666	Movahhed Sadeghi, Soheil Khadirsharbiyani, Ahmed Shokry, Abhishek Shringi, Mostafa Eghbali Zarch and Mahmut Kandemir	Flux-Domain Active Elements for Active Retention in Superconducting Qubits
15	4-THU	10:00 -11:30	703	QTEM	QTEM::1226::666::523	Superconducting Qubit Hardware & Packaging	523	Moritz Singer, Harsh Gupta, Benedikt Schoof, Elena Willinger, Anton Orekhov, Alexandra Schewski, Samuil Kostadinov, Philip Constantin Schneider and Marc Tornow	Substrate-metal interface engineering enhances TaN/Ta thin film superconducting resonator performance
16	4-THU	13:00 -14:30	703	QTEM	QTEM::473::871::223	Ion Traps & Trapped-Ion Platforms	473	Jadwiga Wilkens, Milena Guevara-Bertsch, Marwa Marso, Mederika Zangerl, Florian Girtler, Albert Frisch, Juris Ulmanis, Ingo Roth and Richard Kueng	Local robust shadows on a trapped ion computer --- a case study
17	4-THU	13:00 -14:30	703	QTEM	QTEM::473::871::223	Ion Traps & Trapped-Ion Platforms	871	Mitchell Peaks, Mia Kaarls and Crystal Noel	Low-cost Ultra-low Noise DAC System-on-Module for Scalable Ion-Trap Electrode Control
18	4-THU	13:00 -14:30	703	QTEM	QTEM::473::871::223	Ion Traps & Trapped-Ion Platforms	223	Eike Iseke, Jacob Stupp, Friederike Giebel, Nila Krishnakumar, Konstantin Thronberens, Nora D. Stahr, Md. Masum Billah, Marlon Kuhn and Christian Ospelkaus	Void-Free Through-Glass Vias for Multilayer Surface-Electrode Ion Traps
19	4-THU	15:00 -16:30	703	QTEM	QTEM::1168::1185	Quantum Control, Optimization & Reinforcement Learning II	1168	Temitope Adeniyi and Stefan Andrei	Reinforcement Learning for Quantum Parameter Estimation
20	4-THU	15:00 -16:30	703	QTEM	QTEM::1168::1185	Quantum Control, Optimization & Reinforcement Learning II	1185	Oluwadara Ogunkoya, Tanay Roy, Doga Kurkcuoglu and Silvia Zorzetti	Hardware-aware Optimization of Echoed-Conditional Displacement and Rotation Sequence Parameters
21	4-THU	15:00 -16:30	703	QTEM	QTEM::1168::1185::563	Quantum Control, Optimization & Reinforcement Learning II	563	Charles Cao and Sergei Kalinin	DART-Q: Deadline-Aware Real-Time Feed-Forward for Quantum Control
22	5-FRI	10:00 -11:30	713A	QTEM	QTEM::1245::351::615	Quantum Networking & Quantum Memories	1245	Colin Burdine, Nischal Binod Gautam and Enrique Blair	NEGF Modeling of Impact Ionization in Semiconductor Avalanche Photodiodes for Quantum Networking
23	5-FRI	10:00 -11:30	713A	QTEM	QTEM::1245::351::615	Quantum Networking & Quantum Memories	351	Dheeraj Tukkugudam, Hieu Nguyen, Ethan Hunt, Dillon Horton, Hien Hoang and Tu Nguyen	Optimized Quantum Memory Systems for Quantum Networking: Design and Derivation
24	5-FRI	10:00 -11:30	713A	QTEM	QTEM::1245::351::615	Quantum Networking & Quantum Memories	615	Victor Marcenac, Tommy Nguyen, Julie Chen, Weitao He, Enrique Garcia, Yuyang Han, Bethany E. Matthews, Tiamike Dudley, Andrew Mounce, Kai-Mei C. Fu and Maxwell F. Parsons	Sub-nanosecond control for spin-defect quantum memories with a low-cost, compact FPGA platform
25	5-FRI	13:00 -14:30	713A	QTEM	QTEM::745::678::721	Quantum Sensing & Metrology	745	Atit Pokharel, Ratun Rahman, Thomas Morris and Dinh C. Nguyen	Differential Privacy for Distributed Quantum Sensing via Quantum Noise
26	5-FRI	13:00 -14:30	713A	QTEM	QTEM::745::678::721	Quantum Sensing & Metrology	678	Ratun Rahman, Atit Pokharel and Dinh C. Nguyen	Toward High-Precision Variational Quantum Sensing via Asynchronous Federated Learning
27	5-FRI	13:00 -14:30	713A	QTEM	QTEM::745::678::721	Quantum Sensing & Metrology	721	Priyam Srivastava, Xin Jin, Junyu Liu, Gurudev Dutt, Tom Purdy, Kang Kim and Kaushik Seshadreesan	Variational Joint Magnetometry and Gradiometry on Dipolar Spin Chains
28	5-FRI	15:00 -16:30	713A	QTEM	QTEM::902::816::1271	Quantum Readout & Cryogenic Electronics	902	Chenao Yuan, Nam Truong, Bingjian Du, Dylan Ma, Zhezheng Ren, Justin Llanos, Rubaya Absar, Jonathan Baugh, Lan Wei and Shiyu Su	A Scalable and Cost-Efficient QuBit Readout Circuit with Code-Division Multiplexing in 22nm

QTEM - Quantum Technologies and Systems Engineering

QCE26 Technical Papers Track

29	5-FRI	15:00 -16:30	713A	QTEM	QTEM::902::816::1271	Quantum Readout & Cryogenic Electronics	816	Daryoush Shiri, Likai Yang, Saesun Kim, Mohamed A. Hassan, Philip Krantz and Eric T. Holland	Impact of Pump Phase-Noise on Josephson Traveling-Wave Parametric Amplifiers
30	5-FRI	15:00 -16:30	713A	QTEM	QTEM::902::816::1271	Quantum Readout & Cryogenic Electronics	1271	Yichen Shen, Antoine Henry and Avik Dutt	Quantum Squeezing and Bright Entangled Frequency Combs from a Nanophotonic Chip through Self-homodyning

QGDD - Quantum-GenAI Co-Design Co-Discovery
QCE26 Technical Papers Track

v123	Day	Time	Room	Track	QCE-ID	Session Name	ECID	Paper Authors	Paper Title
1	3-WED	10:00 -11:30	605	QGDD	QGDD::311::339::996	AI-Driven Quantum Circuit Design, Reasoning & Simulation	311	Matthew Tremba, Paul Hovland and Ji Liu	Build Your Own Adder: AI-Driven Discovery of Quantum Prefix Tree Adder Designs
2	3-WED	10:00 -11:30	605	QGDD	QGDD::311::339::996	AI-Driven Quantum Circuit Design, Reasoning & Simulation	339	Sean Feeney, Pooja Rao, Andreas Kiappenecker, Reuben Tate, Yuri Alexeev, Stefano Mensa, Elica Kyoseva and Stephan Eidenbenz	SCALAR: A Neurosymbolic Framework for Automated Conjecture and Reasoning in Quantum Circuit Analysis
3	3-WED	10:00 -11:30	605	QGDD	QGDD::311::339::996	AI-Driven Quantum Circuit Design, Reasoning & Simulation	996	Shamminuj Aktar, Reuben Tate and Stephan Eidenbenz	Structure-Aware Transformers for Learning Near-Optimal Trotter Orderings with System-Size Generalization in 1D Heisenberg Hamiltonians
4	4-THU	10:00 -11:30	705	QGDD	QGDD::947::156::512	LLMs for Quantum Programming & Code Synthesis	947	Minghao Shao, Nuhaila Innan, Hariharan Janardhanan, Muhammad Kashif, Alberto Marchisio and Muhammad Shafique	PennySynth: RAG-Driven Data Synthesis for Automated Quantum Code Generation
5	4-THU	10:00 -11:30	705	QGDD	QGDD::947::156::512	LLMs for Quantum Programming & Code Synthesis	156	Zhenxiao Fu, Lei Jiang, Yilun Xu, Gang Huang and Fan Chen	QAgent: An LLM-based Multi-Agent System for Autonomous OpenQASM Programming
6	4-THU	10:00 -11:30	705	QGDD	QGDD::947::156::512	LLMs for Quantum Programming & Code Synthesis	512	Sharmin Afrose, Ethan Seefried, Nicholas Stapleton, Vicente Leyton Ortega, Travis S Humble and Tirthankar Ghosal	Quantum Coding Assistants: A Survey of LLM-Based Approaches for Quantum Program Synthesis
7	4-THU	13:00 -14:30	705	QGDD	QGDD::265::1180::238	AI-Guided Quantum Optimization & Autonomous Research	265	Seongmin Kim, Abhinav Rijal, Yuri Alexeev, Nora Bauer, Martin Roetteler, Mina Yoon, George Siopsis and In-Saeng Suh	DQAOA-GPT: AI-Accelerated Distributed Quantum Optimization for Combinatorial Problems
8	4-THU	13:00 -14:30	705	QGDD	QGDD::265::1180::238	AI-Guided Quantum Optimization & Autonomous Research	1180	Luis Mantilla Calderon, Jérôme F. Gonthier, Ignacio Gustin, Varinia Bernales and Alan Aspuru-Guzik	Optimizing ground state preparation protocols with autoresearch
9	4-THU	13:00 -14:30	705	QGDD	QGDD::265::1180::238	AI-Guided Quantum Optimization & Autonomous Research	238	Monit Sharma and Hoong Chuin Lau	AutoQResearch: LLM-Guided Closed-Loop Policy Search for Adaptive Variational Quantum Optimization
10	4-THU	15:00 -16:30	705	QGDD	QGDD::496::826::1259	Quantum AI Representations & Generative Models	496	Rogério Feris, Yunchao Liu, Pengyuan Li, Hang Hua and David Kremer	Aligning Quantum Operators with Large Language Models
11	4-THU	15:00 -16:30	705	QGDD	QGDD::496::826::1259	Quantum AI Representations & Generative Models	826	Markus Baumann, Daniel Hein, Steffen Udluft, Tobias Rohe, Claudia Linnhoff-Popien and Jonas Stein	Parity Supervision as a Driver of Generalization in Quantum Generative Modeling
12	4-THU	15:00 -16:30	705	QGDD	QGDD::496::826::1259	Quantum AI Representations & Generative Models	1259	Yu-Cheng Lin, Yu-Chao Hsu, I-Shan Tsai, Chun-Hua Lin, Kuo-Chung Peng, Jiun-Cheng Jiang, Yun-Yuan Wang, Tzung-Chi Huang, Tai-Yue Li, Kuan-Cheng Chen, Samuel Yen-Chi Chen and Nan-Yow Chen	Generative Quantum-inspired Kolmogorov-Arnold Eigensolver

QAPP - Quantum Applications

QCE26 Technical Papers Tracks

v123	Day	Time	Room	Track	QCE-ID	Session Name	ECID	Paper Authors	Paper Title
1	0-SUN	10:00 -11:30	701A	QAPP	QAPP::304::481::1253	Quantum Circuits, Compilation & State Preparation	304	David Quiroga, Hannes Leipold and Bibhas Adhikari	Qvine: Vine Structured Quantum Circuits for Loading High Dimensional Distributions
2	0-SUN	10:00 -11:30	701A	QAPP	QAPP::304::481::1253	Quantum Circuits, Compilation & State Preparation	481	Sahan Sanjaya, Hari Krishna Parvatham, Emma Andrews and Prabhat Mishra	Controlled Steering-Based State Preparation for Adversarial-Robust Quantum Machine Learning
3	0-SUN	10:00 -11:30	701A	QAPP	QAPP::304::481::1253	Quantum Circuits, Compilation & State Preparation	1253	Kazuma Watanabe, Hideaki Kawaguchi, Shin Nishio and Takahiko Satoh	Digital Annealer-Assisted Accuracy-First Quantum Circuit Transpilation with Integrated QUBO Mapping and Routing
4	0-SUN	15:00 -16:30	701A	QAPP	QAPP::904::935::1258	Quantum Hardware, Architecture & QRAM	904	Erik Viirre and Mingxiong Huang	RF SQUID Magnetometer Array with Integrated Quantum Computing for Brain Imaging: Design Requirements and Technical Challenges
5	0-SUN	15:00 -16:30	701A	QAPP	QAPP::904::935::1258	Quantum Hardware, Architecture & QRAM	935	Ansh Singal and Kaitlin N. Smith	VarSEQ: Variable Strength Error-corrected QRAM
6	0-SUN	15:00 -16:30	701A	QAPP	QAPP::904::935::1258	Quantum Hardware, Architecture & QRAM	1258	Vincenzo Lipardi, Domenica Dibenedetto, Georgios Stamoulis and Mark H.M. Winands	Magic-Informed Quantum Architecture Search
7	0-SUN	13:00 -14:30	701A	QAPP	QAPP::202::1129	Quantum Error Mitigation, Fidelity & Verification	202	Philipp Aumann, Michael Fellner, David Alber, Max Cykier, Christoph Fleckenstein, Roeland ter Hoeven, Leo Stenzel, Riccardo Valencia Tortora and Wolfgang Lechner	
8	0-SUN	13:00 -14:30	701A	QAPP	QAPP::202::1129	Quantum Error Mitigation, Fidelity & Verification	1129	Phat Thanh Tran and Xiuwen Liu	LUCYD-Q: A Hamming-Aware Deconvolution Network for Quantum Error Mitigation
9	1-MON	15:00 -16:30	716A	QAPP	QAPP::627::1112::1188	Graphs & Quantum Data Structures	582	Jishnu Mahmud, John Winship, Tom Lash, James Ostrowski and Rebekah Herman	Explicit Block Encoding of Difference-of-Gaussian Operators on a Periodic Grid
10	1-MON	15:00 -16:30	716A	QAPP	QAPP::627::1112::1188	Graphs & Quantum Data Structures	849	Bibhas Adhikari	Quantum embedding of graphs for subgraph counting
11	1-MON	15:00 -16:30	716A	QAPP	QAPP::627::1112::1188	Graphs & Quantum Data Structures	1160	Takeshi Yamane and Naoki Yamamoto	Polarization-Resolved Electron Paramagnetic Resonance for Direct Reconstruction of the Quantum Geometric Tensor
12	1-MON	15:00 -16:30	716B	QAPP	QAPP::582::849::1160	Quantum Learning for Healthcare & Medical Imaging	627	Mohammad Chalhoub, Mahdi Chehimi, Laia Domingo, Omar Alhussein, Ahmad Farouk and Saif Al-Kuwari	Superpixel-Based QUBO for Scalable Quantum-Enhanced Medical Image Segmentation
13	1-MON	15:00 -16:30	716B	QAPP	QAPP::582::849::1160	Quantum Learning for Healthcare & Medical Imaging	1112	Isabella Florez, Ahmed Farhat, James Le Houx, Edoardo Altamura and Gianluca Tozzi	Quantum Kernel Support Vector Machines for Trabecular Bone Classification: Comparing Feature Reduction Strategies on Synthetic Micro-CT Data
14	1-MON	15:00 -16:30	716B	QAPP	QAPP::582::849::1160	Quantum Learning for Healthcare & Medical Imaging	1188	Satish Wagle, Saroj Baral, Clifford Jones, Khem Poudel and Hanna Terletska	Quantum Computing for Healthcare: A Cross-Domain Taxonomy of Algorithms, Hardware Readiness, and Open Challenges
15	2-TUE	10:00 -11:30	802A/B	QAPP	QAPP::257::600::914	Financial Optimization I – Portfolio Optimization	257	Fabio Vitale, Francesca Cibrario, Davide Veronelli, Chiara Vercellino, Valeria Zaffaroni, Giacomo Vitali, Gregorio Antonio Polito, Emanuele Dri, Carla Monferrato, Olivier Terzo, Davide Corbelleto and Marco Bianchetti	Quantum-Inspired Risk Portfolio Optimization via Exponential Tilting
16	2-TUE	10:00 -11:30	802A/B	QAPP	QAPP::257::600::914	Financial Optimization I – Portfolio Optimization	600	Jose Falla and Ilya Safro	Constrained Counterdiabatic Quantum Approximate Optimization Algorithm for Portfolio Optimization
17	2-TUE	10:00 -11:30	802A/B	QAPP	QAPP::257::600::914	Financial Optimization I – Portfolio Optimization	914	Xiaoguang Yang, Menghan Dou and Guoping Guo	A Hybrid Quantum-Classical Coordination Architecture for Portfolio Optimization via Global Context Injection
18	2-TUE	13:00 -14:30	802A/B	QAPP	QAPP::919::920::924	Financial Optimization II – Encoding, Markets & Analytics	919	Takuya Yoshioka, Keita Sasada, Riku Usuki and Keisuke Fujii	Performance Analysis of Pauli Correlation Encoding for Electric Power Portfolio Optimization under Varying Procurement Levels
19	2-TUE	13:00 -14:30	802A/B	QAPP	QAPP::919::920::924	Financial Optimization II – Encoding, Markets & Analytics	920	Rahul Rana, Suman Kumar Roy and M Girish Chandra	Experimental Insights of Pauli Correlation Encoding for Currency Arbitrage on NISQ Hardware
20	2-TUE	13:00 -14:30	802A/B	QAPP	QAPP::919::920::924	Financial Optimization II – Encoding, Markets & Analytics	924	Giuliana Siddi Moreau, Maria Laura Clemente, Lorenzo Pisani, Carlo Impagliazzo, Muriel Cабianca, Manuela Profir and Lidia Leoni	Quantum-Assisted Configuration of Renewable Energy Communities via Block Decomposed QUBO Approximation
21	2-TUE	15:00 -16:30	802A/B	QAPP	QAPP::275::292::453	Quantum Optimization for Energy Systems	275	Yusuke Suzuki, Kosuke Yamamoto, Yuki Hayashino, Akihisa Sekiguchi, Mitsuharu Takeori, Noriaki Shimada and Hirotochi Hirai	Quantum-Enhanced Bayesian Optimization: Benchmarks and Application to Semiconductor Manufacturing Processes
22	2-TUE	15:00 -16:30	802A/B	QAPP	QAPP::275::292::453	Quantum Optimization for Energy Systems	292	Chihiro Dogo, Kazuhiro Saito, Masashi Tawada and Nozomu Togawa	Constraint-Driven QUBO for PV 5G Base-Station EMS: Tuning-Free Feasible Optimization
23	2-TUE	15:00 -16:30	802A/B	QAPP	QAPP::275::292::453	Quantum Optimization for Energy Systems	453	Arul Mazumder and Shreyan Mazumder	Depth-Efficient Quantum Topological Data Analysis for Regime-Specific Detection of Financial Stress
24	3-WED	10:00 -11:30	802A/B	QAPP	QAPP::450::961::1118	Quantum Optimization Algorithms & Benchmarks	450	Lennart Binkowski	Practical lower bounds for hybrid quantum interior point methods in linear programming
25	3-WED	10:00 -11:30	802A/B	QAPP	QAPP::450::961::1118	Quantum Optimization Algorithms & Benchmarks	961	Hila Safi, Karen Wintersperger, Oliver von Sicard, Christoph Niedermeier and Wolfgang Maurer	A Reality Check on Quantum Optimisation: Evidence from an Industrial Case Study
26	3-WED	10:00 -11:30	802A/B	QAPP	QAPP::450::961::1118	Quantum Optimization Algorithms & Benchmarks	1118	Alejandro Giraldo Quintero, Daniel Sierra-Sosa and Adel Elmaghraby	Quadratic Speedup on a Spatial Optimization Algorithm: Grover Descent Algorithm
27	3-WED	13:00 -14:30	802A/B	QAPP	QAPP::42::549::901	Quantum Phase Estimation & Hamiltonian Simulation	42	Elijah Pelofske and Stephan Eidenbenz	Numerical Experiments with Parameter Setting of Trotterized Quantum Phase Estimation for Quantum Hamiltonian Ground State Computation
28	3-WED	13:00 -14:30	802A/B	QAPP	QAPP::42::549::901	Quantum Phase Estimation & Hamiltonian Simulation	549	Gautam Sharma, Apurva Tiwari, Niladri Gomes, J. Eric Bracken, Jay Pathak and Jezer Jojo	Hardware Realization of a Hamiltonian Simulation Algorithm for Time-Domain Maxwell's Equations
29	3-WED	13:00 -14:30	802A/B	QAPP	QAPP::42::549::901	Quantum Phase Estimation & Hamiltonian Simulation	901	J��r��mie Messud and Wassil Sennane	Refining Quantum Phase Estimation Precision Conditions on Unitaries for Many-Electron Systems
30	3-WED	15:00 -16:30	802A/B	QAPP	QAPP::283::742::892	Quantum PDEs & Scientific Computing	283	Taito Mizutani, Naoki Ishikawa, Arnaud Lamy, Robert Heath and Koji Ishibashi	Quantum Singular Value Transformation for Solving Partial Differential Equations and Gate Count Analysis
31	3-WED	15:00 -16:30	802A/B	QAPP	QAPP::283::742::892	Quantum PDEs & Scientific Computing	742	Sayonee Ray, Jezer Jojo, Jason Iaconis, Abeynaya Gnanasekaran, Apurva Tiwari, Martin Roetteler, Chris Hill and Jay Pathak	Quantum Lattice Boltzmann Solutions for Transport under 3D Spatially Varying Advection on Trapped Ion Hardware
32	3-WED	15:00 -16:30	802A/B	QAPP	QAPP::283::742::892	Quantum PDEs & Scientific Computing	892	Mahmoud Elkarargy, Abdelaziz Rahwan, Abdelrahman Elsayed and Forat Hatem	Quantum PDE Solvers in Practice: Application-Driven Benchmarking of the Heat Equation

QAPP - Quantum Applications

QCE26 Technical Papers Tracks

33	4-THU	10:00 -11:30	802A/B	QAPP	QAPP::486::524::546	Quantum Machine Learning Applications	486	Natansh Mathur, Panagiotis Barkoutsos, Masako Yamada, Iordanis Kerenidis and Martin Roetteler	Scalable on-hardware training of quantum neural networks and application to clinical data imputation
34	4-THU	10:00 -11:30	802A/B	QAPP	QAPP::486::524::546	Quantum Machine Learning Applications	524	Oliver Knitter, Sang Hyub Kim, Maximilian Wurzer, Jonathan Mei, Claudio Grotto, Karen Horovitz, Chi Chen, Masako Yamada, Frederik Flöther and Martin Roetteler	Measuring Accuracy and Energy-to-Solution of Quantum Fine-Tuning of Foundational AI Models
35	4-THU	10:00 -11:30	802A/B	QAPP	QAPP::486::524::546	Quantum Machine Learning Applications	546	Jingyan Xu, Yuanhong Wu, Samuel Yen-Chi Chen, Sean Patten, Christina Schweikert and D. Frank Hsu	Diversity-Aware Ensemble of Quantum-Classical Models via Combinatorial Fusion
36	4-THU	13:00 -14:30	802A/B	QAPP	QAPP::320::325::439	Quantum Computing Applications & Case Studies	320	Tristan Zaborniak and Vikram Mulligan	Playing Dice with the Universe: Programming Quantum Computers to Play Traditional Games
37	4-THU	13:00 -14:30	802A/B	QAPP	QAPP::320::325::439	Quantum Computing Applications & Case Studies	325	Paul-Niklas Kandora, Adrian Asmund Fessler, Robert Fabian Lindermann, Phil Arnold, Andreas Hempel and Steffen Rebennack	From Cables to Qubits: A Decomposed Variational Quantum Optimization Pipeline
38	4-THU	13:00 -14:30	802A/B	QAPP	QAPP::320::325::439	Quantum Computing Applications & Case Studies	439	Karie Nicholas and Vikram Mulligan	Entangled happily ever after: Wedding reception seating mapped to classical and quantum optimizers
39	4-THU	15:00 -16:30	802A/B	QAPP	QAPP::233::753::1158	QUBO & Discrete Optimization	233	Laurin Demmler and Maximilian Hess	A Nested Amplitude Amplification protocol for the binary knapsack problem
40	4-THU	15:00 -16:30	802A/B	QAPP	QAPP::233::753::1158	QUBO & Discrete Optimization	753	Shun Furusawa, Chihiro Dogo, Kazuhiro Saito, Yuya Seki, Shuta Kikuchi and Shu Tanaka	A Rotation-Aligned Time-Series Calibration for Factorization Machine with Quadratic-Optimization Annealing in Dynamic Combinatorial Optimization
41	4-THU	15:00 -16:30	802A/B	QAPP	QAPP::233::753::1158	QUBO & Discrete Optimization	1158	Sutapa Samanta, Dagen Wang and Andras Ferenczi	Approximate QUBO with classical post-selection for solving the 0/1 knapsack problem
42	5-FRI	10:00 -11:30	604	QAPP	QAPP::228::505::1251	Quantum Chemistry & Molecular Science	228	Francesco Micucci, Matteo Barbieri, Gabriella Bettonte, Domenico Bonanni, Anita Camillini, Anna Fava, Daniele Gregori, Andrea R. Beccari and Gianluca Palermo	A Co-Design Approach to Physically-Informed Molecular Docking for Quantum Annealers
43	5-FRI	10:00 -11:30	604	QAPP	QAPP::228::505::1251	Quantum Chemistry & Molecular Science	505	Yuhei Tachi, Akihiko Arakawa, Taisei Osawa, Masayoshi Terabe and Kenji Sugisaki	
44	5-FRI	10:00 -11:30	604	QAPP	QAPP::228::505::1251	Quantum Chemistry & Molecular Science	1251	Kyungmin Kim, Eyuel Elala, Hoilim Kim, Pilsun Yoo, Doo Hyung Kang, Pratanphorn Nakliang and June-Koo Kevin Rhee	Quantum Computation of Molecular Excited States Using Handover Iterative Variational Quantum Deflation
45	5-FRI	10:00 -11:30	802A/B	QAPP	QAPP::500::1026::1225	Quantum Combinatorial Optimization & Scheduling	500	Dominik Soós, Marc Paterno, John Stenge and Nikos Chrisochoides	Distributed Quantum-Enhanced Optimization: A Topographical Preconditioning Approach for High-Dimensional Search
46	5-FRI	10:00 -11:30	802A/B	QAPP	QAPP::500::1026::1225	Quantum Combinatorial Optimization & Scheduling	1026	Van-Quang-Huy Nguyen, Hoang-Quan Nguyen, Samee U. Khan, Ilya Safro and Khoa Luu	QUACOD: Quantum Optimization via Coordinate Descent for Scalable Drone Scheduling
47	5-FRI	10:00 -11:30	802A/B	QAPP	QAPP::500::1026::1225	Quantum Combinatorial Optimization & Scheduling	1225	Farzan Moosavi and Bilal Farooq	RL-Guided Quantum-ALNS for Constrained VRP
48	5-FRI	13:00 -14:30	604	QAPP	QAPP::270::769::1207	Quantum Simulation of Physical Systems	270	Takanori Nakao, Yuki Furue, Noriniko Takanashi, Yutaka Takita, Marcos Diez Garcia, Antonio Márquez Romero, Jofre Vallès-Muns, Sergio Sanchez-Ramirez and Artur Garcia-Saez	Tensor Network Simulation using matrix product state for a 100-Qubit Variational Quantum Eigensolver in Tetraethyl Orthosilicate Molecule
49	5-FRI	13:00 -14:30	604	QAPP	QAPP::270::769::1207	Quantum Simulation of Physical Systems	769	Ssu-Yi Chen, Bo-Hung Chen, Dah-Wei Chiou and Jie-Hong Roland Jiang	Adiabatic Quantum Simulation of the Topological Su–Schrieffer–Heeger–Hubbard Model
50	5-FRI	13:00 -14:30	604	QAPP	QAPP::270::769::1207	Quantum Simulation of Physical Systems	1207	Takaki Akiba and Youhi Morii	GPU-Based Implementations of QPE and HHL with cuQuantum for Practical Quantum Emulation
51	5-FRI	13:00 -14:30	802A/B	QAPP	QAPP::492::664::786	Quantum Reservoir Computing & Dynamical Learning	492	Stefan Milenkovic, Maria Letizia Marini, James Brown, Kenneth Heitritter, Tarini Shekhar Hardikar, Wesley Ta and Matteo Ceccarelli	Accelerating dipole predictions with Quantum Reservoir Computing
52	5-FRI	13:00 -14:30	802A/B	QAPP	QAPP::492::664::786	Quantum Reservoir Computing & Dynamical Learning	664	Tao Liu, Ge He, Dongyu Liang and Wujie Wen	Evaluating Hybrid Quantum-Classical Models for Reduced-Order Brain Deformation Dynamics
53	5-FRI	13:00 -14:30	802A/B	QAPP	QAPP::492::664::786	Quantum Reservoir Computing & Dynamical Learning	786	Annemarie Wolff, Guillaume Dumas, Filippo Utro, Kahn Rhrissorrakrai, Kathleen Hamilton and Laxmi Parida	A Quantum Reservoir for Neurodynamical Forecasting
54	5-FRI	15:00 -16:30	604	QAPP	QAPP::819::884::1027	Quantum Time-Series Forecasting & Industrial AI	819	Matthew Sato and Kincho Law	LPBF Melt Pool Prediction Using Quantum Computers and Machine Learning
55	5-FRI	15:00 -16:30	604	QAPP	QAPP::819::884::1027	Quantum Time-Series Forecasting & Industrial AI	884	Paul Griffin, Hari Hara Suthan Chittoor, Chengran Yang, Jianjun Chen, Rakesh Achutha, Catalin Burlacu, Randall Duran, Ariel Neufeld, Jayne Thompson and Mile Gu	Quantum-Enhanced Modelling and Sampling of Financial Time-Series Data for Rare Event Forecasting
56	5-FRI	15:00 -16:30	604	QAPP	QAPP::819::884::1027	Quantum Time-Series Forecasting & Industrial AI	1027	Hasan Almatrouk, Daniel Fadi Feghali, Kin Tung Michael Ho, Kuan-Cheng Chen, Lily Lee, Kyriakos Kandris, Evina Katsou and Po-Heng Lee	Quantum Kernel Long Short-Term Memory for Multi-Step Greenhouse Gas Forecasting in Wastewater Treatment Plants
57	5-FRI	15:00 -16:30	802A/B	QAPP	QAPP::487::503::773	Quantum Security & Emerging Architectures	487	James Brown, Jason Iaconis, Yuri Alexeev, Linta Joseph, Spencer Churchill, Kenneth Heitritter, William Aguilar-Calvo, Martin Roetteler and Martin Suchara	Mid-Circuit Measurements for Clifford Noise Reduction in Hamiltonian Simulations
58	5-FRI	15:00 -16:30	802A/B	QAPP	QAPP::487::503::773	Quantum Security & Emerging Architectures	503	Walid El Maoquai, Alberto Marchisio and Muhammad Shafique	Q-LEAK: Quantum-Based LEAKage Verification for Side-Channel Countermeasures
59	5-FRI	15:00 -16:30	802A/B	QAPP	QAPP::487::503::773	Quantum Security & Emerging Architectures	773	Shogo Shimada, Rei Sato, Pulak Ranjan Giri and Kazuhiro Saito	Circuit-Level Analysis of a Quantum Walk-based Hash Architecture for Proof of Quantum Work